CLIMATE SMART AGRICULTURAL TECHNOLOGIES, INNOVATIONS AND MANAGEMENT PRACTICES FOR BANANA VALUE CHAIN

Training of Trainers' Manual

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Foreword

Kenya Climate-Smart Agriculture Project (KCSAP) tasked the Kenya Agricultural & Livestock Research Organization (KALRO) with the implementation of the project Component 2, on 'Strengthening Climate-Smart Agricultural Research and Seed Systems'. The component activities are geared towards the development, validation, adoption and delivery of context specific climate smart agriculture (CSA) technologies, innovation and management practices (TIMPS). The other responsibility was development of sustainable seed production and distribution systems for priority value chains to enhance availability and access to seed, breeds and fingerlings by target beneficiaries under Components 1 (Up scaling Climate-Smart Agricultural Practices). Against this background, KALRO and her NARS partners have developed, validated and availed CSA TIMPS for dissemination and adoption. The TIMPS have further been unpacked during the development of Training of Trainers (ToT) Manuals for use in training public and private extension service providers and lead farmers.

The ToT Manuals are instructional guides to be used for teaching and learning step-by-step procedures of implementing CSA innovations for each of the 13 value chains being addressed. The training content is drawn from the CSA TIMPS that support respective value chains. The content are arranged in progressive modules supported by extensive information from research information and background data drawn from the TIMPS. Their relevance are based on the needs teased out of the value chains and the project objectives. The ToT Manuals training design takes into consideration the delivery system, the partners and their roles, the duration of training and logical flow of the sessions. Similar content requiring similar delivery systems are grouped together while the roles of the partners are tapped in the training and planning of the training sessions.

The Manual is divided into modules, which have a uniform outline that ensures every aspect of the TIMPs are fully covered in way that the trainees can absorb and relate to. Various delivery methods are deployed and where possible demonstrations and practical work are incorporated to enable the trainees learn by participating in the actual field activities. Furthermore, to ensure that the training across various groups is standardized, trainers guidelines, detailed descriptions of the trainees, program, training methods and a training evaluation have been provided in the manual. Adhering to these guidelines, therefore, enables possibility to replicate the training in several locations without loss of details regardless of whether conducted by different trainers.

It is highly advised that the ToT Manuals should be used in conjunction with the respective value chains' TIMPs documents and facts sheets in order to provide valuable resource for both public and private extension service providers. The use of this Manual is expected to spur increased productivity and resilience by farmers, while mitigating climate change impacts in the value chains to deliver the envisaged 'Triple Wins'.

I am greatly indebted to the value chain leaders and all those who participated in the preparation of the Manual, which is expected to herald a new way of delivering training content in a changing agricultural environment.

Eliud K Kireger, PhD, OGW

Director General, KALRO

Preface

The Kenya Climate-Smart Agriculture Project (KCSAP) is a Government of Kenya project with support from both the World Bank and the government. It is a five-year project implemented in 24 counties, mainly in the arid and semi-arid lands (ASALs), at a cost of Ksh. 25B. The project development objective (PDO) is “to increase agricultural productivity and build resilience to climate change risks in the targeted smallholder farming and pastoral communities, and in the event of an Eligible Crisis or Emergency, to provide immediate and effective response.” This objective is to be achieved through the implementation of five key components, which are 1) Up scaling Climate-Smart Agricultural Practices, 2) Strengthening Climate-Smart Agricultural Research and Seed Systems, 3) Supporting Agro-weather, Market, Climate, and Advisory Services, 4) Project Coordination and Management and 5) Contingency Emergency Response.

Component 1 involves facilitating the empowering of farmers and communities to adopt technologies, innovations and management practices (TIMPs) to achieve the Climate Smart Agriculture (CSA) triple-wins of; increased productivity, enhanced resilience (adaptation), and reduced Greenhouse gas (GHG) emissions (mitigation). Component 2 is charged with the responsibility of providing the TIMPs. Therefore, it supports the development, validation, and adoption of context specific CSA TIMPS to target beneficiaries under Components 1 and 3 as well as development of sustainable seed production and distribution systems.

To catalyze uptake of TIMPs, Kenya Agricultural & Livestock Research Organization (KALRO) in conjunction with partners in the National Agricultural Research Systems (NARS) and Consultative Group for International Agricultural Research (CGIAR) compiled inventories of TIMPs for each of the 13 prioritized value chains (cassava, green grams, sorghum, millet, pigeon peas, bananas, tomatoes, potatoes, apiculture, indigenous chicken (meat and eggs), dairy (cattle and camel), red meat (cattle, sheep and goats) and aquaculture and 3 cross cutting value chains (natural resource management, pastures and fodder and animal health). The TIMPs were categorized into those ready for upscaling, those that needed validation and gaps that required further research. Training of Trainers’ (ToT) manuals focusing on TIMPs that are ready upscaling for each of the value chains were subsequently developed and form the basis of training county extension staff, service providers and lead farmers. They are in turn expected to cascade this training to beneficiaries in the targeted smallholder farming, agro-pastral and pastoral communities in the 24 project counties of Marsabit, Isiolo, Tana River, Garissa, Wajir, Mandera, West Pokot, Baringo, Laikipia, Machakos, Nyeri, Tharaka Nithi, Lamu, Taita Taveta, Kajiado, Busia, Siaya, Nyandarua, Bomet, Kericho, Kakamega, Uasin Gishu, Elgeyo Marakwet and Kisumu.

KALRO having the mandate of implementing of activities under Component 2, has been instrumental in using its information resources and those of partners and collaborators to come up with the inventories of TIMPs and corresponding ToT Manuals. The use of these information resources coupled with the accompanying training and the contribution of the other project components, will go a long way in enabling the KCSAP to meet its development objective.

The National Project Coordination Unit is grateful to all who participated in the development and production of this Training of Trainers' Manual for *Climate Smart Agricultural Technologies, Innovations and Management Practices for Banana Value Chain*. It is my hope that counties and other users will put this resource to good use as they transform and reorient their agricultural systems to make them more productive and resilient while minimizing GHG emissions under the new realities of a changing climate.

Francis Muthami

National Project Coordinator
Kenya Climate-Smart Agriculture Project

LIST OF ABBREVIATIONS AND ACRONYMS

BXW	Banana Xanthomonas Wilt
CSA	Climate Smart Agriculture
CCT	County Coordination Teams
CTT	Core Team of Trainers
EFMIS	Enhanced Fish Market Information Systems
ESP	Economic Stimulus Program
FAO	Food and Agriculture Organization
FFBS	Farmer Field and Business School
FFS	Farmer Field School
FGDs	Focus Group Discussions
GAP	Good Agricultural Practices
GDP	Gross Domestic Product
KALRO	Kenya Agriculture and Livestock Research Organization
KCSAP	Kenya Climate Smart Agriculture
LF	Lead Farmer
MTP	Medium Term Plan
MOALF&I	Ministry of Agriculture, Livestock, Fisheries and Irrigation
NEMA	National Environmental Management Authority
NGO	Non-Governmental Organizations
TIMPS	Technology Innovation Management Practices
TNA	Training Needs Assessment
ToT	Training of Trainers
SDGs	Sustainable Development Goals

INTRODUCTION

About this manual

This training of Trainers' Manual consists of two parts; namely Part I and Part II. Part I comprises notes for the facilitators while Part II is made up of the training modules in the value chain. It is expected that the organizers of the training will make prior arrangements with the facilitators to identify the training venues and prepare demonstrations sites in advance in order to make all the training sessions a success.

PART I

This part consists of four sections including the background, module training content, training design and facilitator guidelines.

SECTION 1 BACKGROUND

1.1 Banana industry in Kenya

Banana (*Musa sp*) is one of the most important fruit crops, accounting for 32 % of the total value of fruits and hence ranks first among fruit crops in Kenya. The major banana producing counties are: Meru 17%, Kirinyaga 11%, Muranga 9 %, Kisii 8%, Tharaka 6%, Kiambu 5% and Taita Taveta 5%. The main varieties grown by small-scale farmers include: dessert banana cultivars (Grand Nain, Gross Mitchel, Williams hybrid, Valery, Chinese Cavendish, Giant Cavendish, Dwarf Cavendish and Apple), and cooking cultivars (Gradi, Shisikame, Mutahato, Uganda green and Ng'ombe) and multipurpose cultivars (such as Muraru and Gold finger). Banana is as a major source of food, livestock feed and cash income in most parts of the country. The sub sector employs about 80 % of the rural community and contributes over KES 16.8 billion to the Kenyan economy. Hence plays a crucial role in terms of contribution to food security, poverty eradication, and economic development.

1.2 Role of Banana in foood and nutrition security

Despite its great potential in contributing to food, economic and nutritional security, small holder farmers have not realized the benefits of banana. For instance under research, the potential of banana yields range between 30-40 tons/ha, however smallholder farmers realize less than 10 tons/ha. This is attributed to several constraints namely: inadequate information on climate smart agricultural practices, lack of information on the source and availability of clean high yielding banana planting materials, inadequate knowledge on establishment and management of a hardening nursery and macro-propagation, inadequate knowledge on orchard establishment and management, erratic rainfall due to climate change, low soil fertility, lack of information on soil and water management, lack on site-specific information on fertilizer requirements, high cost of farm inputs, high incidence of pests and diseases such as weevil borers, nematodes, Fusarium wilt, Sigatoka and Xanthomonas Wilt (BXW), high postharvest losses of about 40%, disorganized marketing and poor marketing infrastructure (especially roads and grading sheds) as well as low value addition mainly due to low quality produce, lack of business skills to commercialize banana enterprise, low gender and youth participation in projects, lack of quantitative information on banana sector performance and limited group networks and cohesion among farmers.

1.3 Banana as a Climate Smart Innovation

Banana is one of the best commodity options for food security under the changing climate while delivering co-benefits for environmental sustainability, nutrition and livelihoods. The SDGs and many associated targets are directly relevant for policy making, planning and management for sustainable development of agriculture. However, agriculture faces increasing constraints including declining soil fertility, water scarcity/ drought. These factors combined with other potential impacts of climate change on ecosystems and communities, present significant challenges to the entire sector.

This manual examines the Climate-Smart Agriculture (CSA) concept from the banana perspective. The Kenya Climate Smart Agriculture Project (KCSAP) aims to validate and upscale banana Technologies, Innovations and Management Practices (TIMPs) through (1) improving efficiency in the use of land resources to produce bananas; (2) maintaining the resilience of agricultural systems and the dependent communities; and (3) gaining an understanding on how to reduce the vulnerability of communities negatively impacted by climate change in Kenya.

SECTION 2 TRAINING CONTENT

2.1 Orientation of the Modules

The first part of this section is about the training content or modules. It outlines the orientation and outline of the 13 modules which have 79 sessions. The first 5 modules provide details on training in facilitation skills, the FFS as the extension methodology, participatory monitoring and evaluation ending up scaling and sustainability. The following 6 modules cover Good Agricultural Practices (GAP) that includes farming and market assessment.

The 11 modules are positioned to ensure Good Agricultural Practices (GAP) are adopted to improve productivity through improved Banana value chain competitiveness in a market led production. During the Training Needs Assessment (TNA), farmer's priority needs in knowledge were production, postharvest, marketing and adoption of GAP for increased productivity.

2 .2 Module Outline

Each of the 9 modules has a similar outline consisting of 8 parts. These parts are:

- **Introduction to the module** – context and background to training needs, knowledge and skill gaps being addressed.
- **Module learning outcomes** – what trainees are expected to learn.
- **Module target group** – trainee categories.
- **Module users** – facilitators.
- **Module duration** – The minimum number of hours of exposure to materials.
- **Module summary** – The sequence of sessions, training methods, materials and duration.
- **Facilitators guideline** – detailed sessions, training methods, materials and session guides.
- **Participant's handouts** – detailed training notes and reference materials for trainees.

The outline for each of the 11 modules is presented in Table 1:

Table 1: Summary of outline for 13 modules of the Banana value chain

No.	Module name	Need addressed	Expected training outcomes [Summarized from module learning outcomes]	M&E Indicators and Means of Verification (MoV) [Training focus]	Duration
1	Climate Smart Agriculture Practices	• Rapid growth in food demand intensified by the effects of climate change on agricultural systems, including crops, livestock, forestry and fisheries.	• Impacts of climate change and the linkages between climate, agriculture and food security understood	• Training hours • The no. of people trained • Person days of training • Number of trainers trained through ToT activities • No of TIMPs promoted in ToT trainings sessions • Attendance lists (Disaggregated by gender) • Trainers notes • PowerPoint presentation • Participant evaluation • Training report • Demo materials for practical	3 hours

2	Farmer field and business school (FFBS) approach in banana production	Lack of knowledge on Technologies, Innovations and management Practice (TIMPs)	• Have a clear understanding of Farmer Field and Business School approach and be able to differentiate between teaching and facilitating. • Gain practical skills that enables them to to facilitate a participatory learning session. • Gain knowledge and analytical skills to design simple experiments to test and select the best solution to identified (problem) challenges (TIMPs). • Be able tofacilitate the shift from the traditional focus to improving productivity towards farming business proposition.		4 hours 40 Minutes
3	Banana production niche and climatic requirements	Lack of knowledge on areas suitable for banana production and their ecological characteristics	Knowledge of Banana production niche and appropriate climatic conditions for optimal yield demonstrated		4 Hours

4	Improved banana varieties	• Low banana crop yields and productivity from current farmer planted varieties and cultivars • Lack of knowledge on suitable varieties for various banana production niches	• Banana varieties suitable for different AEZ’s for increased productivity identified • Characteristics of different banana varieties identified		4 hours 20 minutes
5	Banana Propagation (Seed Systems)	• Lack of information on the source and availability of clean planting materials • Inadequate knowledge on establishment and management of a hardening nursery and macro-propagation unit	• Sources of clean Explants and TC plantlets for supply to TC labs and Hardening nurseries identified • Knowledge and skills in setting up and maintaining hardening nurseries and macro-propagation units learnt and demonstrated		4 hours 30 minutes
6	Good Agronomic Practices (Orchard establishment and management)	• Lack of knowledge and skills on appropriate orchard establishment and management practices	• Knowledge and skills on orchard establishment and management learnt • Suitable intercrops for bananas identified and their cross management documented		5 hours 30 minutes

7	Integrated soil fertility and water management practices for banana production	• Inadequate knowledge on appropriate soil and water management for optimized banana production	• Knowledge of Soil fertility and nutrient management practices demonstrated • Water management and crop irrigation needs established		6 hours
8	Crop Health (i) Banana pests and their control	• Loss of yields and income due to increased incidences and severity of major and emerging banana pests • Lack of knowledge and skills on pest identification and control in banana orchards Lack of knowledge on available ITK's for pest management • Lack of knowledge on safe use of pesticides	• Surveillance mechanisms for emerging and major banana pests identified and learnt • Major banana pests identified and their economic importance learnt and demonstrated • Management of major banana pests learnt and documented • Available ITK's for pest management documented • Knowledge and skills on safe use of pesticides learnt		6 hours

	(ii) Banana diseases and their control	• Increased incidences and severity of major and emerging banana diseases that lead to yield and income losses • Lack of knowledge and skills on disease identification and control in banana orchards • Lack of knowledge on available ITK's for disease management	• Surveillance mechanisms for emerging and major banana diseases identified and learnt • Major banana diseases identified and their economic importance learnt and demonstrated • Management of major banana diseases learnt and documented • Available ITK's for disease management documented		
	(iii) Banana weeds management	• Lack of knowledge and skills on weed identification and control in banana orchards	• Major banana weeds identified and their economic importance learnt and demonstrated		
	(iv) Pollination management in bananas				

9	Postharvest management	• Inadequate knowledge and skills on appropriate post-harvest technologies leads to high losses in bananas • Lack of information on available ITK's for banana Post-harvest management	• Appropriate post-harvest technologies identified, learnt and documented • Available ITK's for banana post-harvest management learnt and documented		5 hours 30 minutes
10	Value addition	• Lack of information on nutrient profile of the various banana varieties • Lack of information, knowledge and skills in value addition and processing of various banana products	• Nutrient profile of the various banana varieties documented • Enhanced knowledge, information and skills in processing of Juices, wine, dried, composite flour and baked products learnt		5 hours
11	Mechanization of Banana production	• Inadequate knowledge and skills on appropriate mechanization technologies leads to high losses in bananas	• Appropriate mechanization technologies identified, learnt and documented		2 hours

12	Banana farming business and Marketing	Lack of Understanding of Operations of Farm Business	Banana farming business and Marketing		4 hours
13	**Cross Cutting issues**				
	(v) Agricultural Innovation Platform (AIP)		• The attributes of an innovation platform appreciated and applied • Stakeholders mobilized to initiate an Agricultural Innovation Platform • Agricultural Innovation Platforms established, managed and monitored • The Agricultural Innovation Platforms and innovation capacity of AIP actors sustained		2 hours

	(i) Gender, Youth, Vulnerable and Marginalized Groups	• Inadequate understanding of the importance of gender issues in the value chain • Low integration of Women, Men and VMG's in the banana value chain • Inadequate women, Youth and VMG's empowerment	• Importance of gender issues in the value chain learnt and understood • Integration of Women, Men and VMG's in the banana value chain mainstreamed • Women, Youth and VMG's in the value chain empowered • Social cultural factors within the value chain understood		2 hours
	Social and Environmental Performance	• Limited understanding of the social cultural factors within the value chain • Lack of environmental and social safe guards • Inadequate understanding of environmental and social safe guards	• Environmental and social safe guards put in place • Environmental and social safe guards		

	iii Policy	• Limited under-standing of the policy the value chain issues affecting banana value chain	• Policy issues within the value chain understood		2 hours
	TOTAL DURATION				**61 hours**

SECTION 3: THE TRAINING DESIGN

3.1 Delivery system designed for this training consists of two stages:

1. Establishment of a team of facilitators

- A Core Team of Trainers (CTT) will train farmer's trainers (service providers) as facilitators in a 7-day TOT course using this manual.
- Each of the farmer trainers will facilitate farmers to acquire knowledge and skills in facilitating farmer led FFS for Banana through a practical FFS process.
- **2. Up scaling** – from the 1,000 farmers trained, 120 Lead Farmers (LF's) are selected and trained in facilitation skills. In 2019 they facilitated 120 LF led FFBS. By the end of year 1(2019) they will have trained a total of 1800 other farmers in the up-scaling plan.

3.2 Partners and their Roles

The partners envisioned in this training plan are:

1. Core Team of Trainers – master trainers drawn from KALRO, Universities, KEPHIS and Department of Agriculture will facilitate initial training of farmer trainers. They also provide mentorship to farmers' trainers during the first year of LF trainings. They should also be available in the evaluation of the first round of LF trainings.

2. County Government Department of Agriculture –farmer trainers and their supervisors who can be referred to as County Coordination Teams (CCT) to take role of LF trainers, mentors and coordinators at sub county level. They assist FFS's form partnership with stakeholders for sustainability. They should also support LF's form their network.

3. Lead Farmer Networks-association of LFs in the counties to take up farmer trainings and up scaling in the future. The first group of LF in Kericho, Bomet, Nyeri and Tharaka Nithi should visit and learn from Bungoma based Farmer Field Schools Network on how they can start their own.

4. Private Sector Service Providers – inputs suppliers, financial and business development service providers, market players and processors to partner and support growth of individual farmers or farmer groups so as to evolve into sustainable social and commercial entities.

3.3 Training Duration

The proposed initial TOT course for Master trainers for 13 modules in the banana value chain shall take **61 hours** of the training period. This does not include break hours of mid-morning, afternoon and lunch breaks.

3.4 Logic of Design and Flow of Session

The logic of design and flow of each module is that the facilitator, paying attention to the proposed methods and sessions guidelines shall: (1) Introduce the module; (2) Draw out the trainees expectations; (3) Relate trainees' expectations with module objectives or learning outcomes; (4) Explore the concept and content, switching to different methods of delivery of the content (group exercise, brainstorming, excursions, plenary discussions, role plays) as the session progresses; (5) Review the module at the end using participatory approaches; and, (6) Distribute handouts to the participants.

SECTION 4 FACILITATOR GUIDELINES

4.1 Preparation of Training Materials

The training materials suggested require adequate preparations and should be available before the actual training dates. Further:

1. The facilitators should familiarize themselves and internalize the guidelines provided by this manual prior to the training.
2. The stationery required should be available within the training institution three days before the training. These include name tags, writing materials, a paper punch and medium size files for participants' handouts filing.
3. Flip charts and quality felt pens could be used interchangeably with LCD projections. Each participant will require one felt pen while the trainers will require two sets of felt pens.
4. Visual aids like field equipment and tools should also be arranged in time before the sessions start.
5. There should be adequate copies of participants' handouts (one per participant) to be distributed at the end of each session or as may be suitable.
6. Copies of the module training content shall be distributed at the end of each module.

4.2 Preparation of Training Venue and Sites

The training venue will include the training room, field demonstration sites and market areas.

1. **Training Room** – should have adequate space for 25 trainees seated in a semi-circle or U shape arrangement ensuring access and unobstructed view of the front. There should be adequate space for a desk and seats for 3 trainers preferably at the sides or at the back of the training room. There should also be a desk for the trainer, their training materials and LCD projector, a flip charts holder and white wall to act as a projector screen.
2. **Demonstration Site** – Should be within walking distance or less than 10 minutes' drive awaywith well-established banana orchards having different banana varieties.
3. **Market Sites** – these include Banana sale outlets (Markets, collection points, kiosks) and processing sites if any. The operators should be informed in advance about the visits. These should not be very far away preferably less than 10 minutes' drive away.

4.3 The Trainees

The trainees who will participate are extension officers, lead farmers, educators, and researchers with an elaborate training background in extension and advisory services. They will be drawn from public and private sector based on considerable experience in training farmers but with minimal facilitative advisory or technology transfer approaches.

The trainer should, therefore, act more of a facilitator than a lecturer and draw out and build on the knowledge, skills and experience that they shall bring in. As a golden rule, facilitators **should not lecture** trainees but facilitate and listen and let them feel like equals amongst each other and the CTT team members.

4.4 Training Program

The training proposed program consists of the actual training modules. Health breaks should be considered when drawing the training program. It should preferably be based on the outline presented in **Annex 1** to allow the flow of ideas and topics. However, should the situation demand, the sequence and day of coverage for whole or parts of the modules can be modified to suit emerging requirements. The training program assumes that the trainees report on a Sunday evening as the first day and leave seven days later.

4.5 Training Methods

The training methods proposed for each session are suitable for adult learners and appropriate for addressing knowledge, skills and attitudes of the trainees. The choice of the methods has been informed by the competency issues being addressed, available time and experiences of the author of this manual. Depending on time available, the facilitator may modify these training methods but as a golden rule no presentation by the facilitator should take more than **30 minutes** continuously; but should be separated by the other participatory training methods. The list of available training methods is presented in Table 2.

Table 2: Description of training methods during TOT trainings

Training Method	Description of Method
Plenary presentations	Use of PowerPoint or flip charts and plenary discussions in situations where knowledge and opinion or consensus is required
Group exercises, buzz groups, visits and brainstorming sessions	To be considered where skills are an issue requiring sharing and trying
Role plays and problem-solving exercises	Plenary discussions have been considered as training methods where attitude is an issue
On-farm practical demonstration and exchange visits	To be considered where hands-on practical skills are acquired through sharing and demonstration

4.6 Planning Schedule and Guidance for TOT preparation

While planning for this training, the CTT leader should ensure the following activities outlined in Table 3 are conducted before the training commences.

Table 3: Activities to be done before training

Duration to Training	Activities to be done
Six weeks	Recruit master trainers, compose CTT, identify the practical demonstration sites
Four weeks	Send out invitation letters to trainees and special guests detailing purpose, venue and program. Follow up on demonstration sites. Brief CTT members
Two weeks	Confirm names of trainees; reproduce training materials for facilitators and package, confirm preparedness of the field sites to be visited. Hold briefing of CTT members to finalize training plan. Confirm special guests if any
Four days	Confirm training sites preparedness, prepare sitting arrangements, and brief assistants
One day	Arrange training room furniture, place materials, equipment and stationery on the tables. Arrange for the reception of trainees at residence proposed
On the first day	Arrange for the reception of trainees at the training venue. Ensure climate setting is done before the course is officially opened. This includes: (i) registration, (ii) welcome to the venue by host; (iii) elaborate introduction of CTT and trainees; setting of ground rules; and (iv) formation of groups.

4.7 Evaluation of Training

A half day has been allocated for planning for way forward and evaluation of the TOT on the last day of the training. The evaluation strategy should take two directions. The first being the individual trainees evaluate through evaluation forms without conferring to each other (Table 4). The evaluation forms are then collected and analyzed by the CTT members.

Table 4: Individual Sample Evaluation Form Template

Aspect / Module	Rating		
	Very useful (3 marks)	Useful (2 marks)	Least useful (1 mark)
Climate smart agriculture			
Improved banana varieties			
Propagation			
Orchard establishment and management			
Soil and water management for bananas			

Banana Pests and diseases and their control			
Postharvest handling and Value addition			
Banana business			

The second direction for evaluation is trainee's **group evaluation**. They retreat to one room and elect a chair and a secretary. Ask them to objectively and constructively evaluate the training in about 45 minutes in the absence of the CTT members. They then present their evaluation to the CTT members and as they present, the CTT members should only give points of clarifications if any misunderstanding occurred but should not try to be defensive. The CTT members will then use the two evaluation results to write a report highlighting aspects that went on well and can be replicated, challenges that were encountered, and opportunities for future ToT's improvement.

4.8 Participant's Training Notes and Reference Materials

4.8.1. List of banana publications

The list of banana publications are divided into two. The detailed list of all publications is summarized in Annex 2:

1. Publications for potential new entrants and new banana farmers who have just started in the last 6-12 months with simple take home messages. (to be **Labelled A** in Table format)
2. More advanced level publications which can be understood and used by more established banana farmers. (to be **Labelled B** in Table format)

4.8.2 Guide on the use of the information

The trainers will be advised to issue farmers with at most two publications for each of the training sessions. This is because if they go away with many publications, for example, in one visit they may be overwhelmed with the material load and this may limit knowledge uptake. Also, some will just take away as many as they can if allowed.

The list of all individual publications will be stored and available as electronic copies – mainly PDFs. Service providers are strongly advised to keep these electronic copies on a memory stick, or portable hard drive to enable farmers easily access and if necessary print any of them at a local internet café.

Trainers will be advised to issue a General banana production guide. The manual will be accompanied by other publications e.g. information sheets, brochures, factsheets, posters etc. With subsequent training modules, trainees can develop their collection of publications.

PART II

This part consists of 13 modules namely: Climate Smart Agriculture Practices, Farmer Field and Business Schools (FFBS), Banana production niche and climatic requirements, Banana Improved banana varieties, Banana Propagation, Soil Fertility and water Management, Orchard establishment and management, Crop Health, Postharvest handling and management, Value addition, Mechanization in the banana Value Chain, Banana Business, and Banana Gender Mainstreaming and Social Inclusion.

All the module are divided into the following sub-sections:

1.1 Introduction to the module
1.2 Module learning outcomes
1.3 Module target group
1.4 Module users
1.5 Module duration
1.6 Module summary
1.7 Facilitators guideline
1.8 Participants handouts

MODULE 1
CLIMATE CHANGE AND CLIMATE SMART AGRICULTURE

1.1 Introduction to the module

The impact of climate change and variability on agriculture and food security is of serious concern. Kenya's agricultural production systems are highly impacted due to the low adaptive capacity and the high exposure to climate related risks. The major agricultural activities are prone to risks and uncertainties of nature as affected by climate change either in intensity, scope or frequency. Climate change is expected to modify risks, vulnerabilities and the conditions that shape the resilience of agriculture systems as well as introducing new uncertainties. Adoption of climate smart agriculture (CSA) through application of tools and technologies and effective communications of weather information reduces the negative impacts of climate change and enhances access to food security in a changing environment. Thus, there is need to mainstream suitable climate resilient technologies, innovations and management practices (TIMPS) to increase productivity, resilience to climatic shocks and mitigate the causes of climate change.

1.2 Module Learning Outcomes

By the end of the module the following should be achieved:

- Concept of climate change and variability explained and appreciated.
- Concept of Climate smart agriculture (CSA) defined.
- Impacts of climate change and variability on agriculture and food security identified and explained.
- Projected future climate scenarios and how to manage presented and appreciated.

1.3 Module Target Group

These module targets agricultural extension service providers dealing directly with farmer groups at community level or community facilitators.

1.4 Module Users

This module is intede for use by master trainers who are members of the core team of trainers (CTT) and lead farmers in the bananavalue chain target counties. The trainers using this module should thoroughly farmilialize themselves with participants handouts (Training materials)

1.5 Module Duration

The Module is estimated to take about 3 hours.

1.6 Module Summary

Sessions	Training Methods	Training Materials	Time
1. Introduction to climate change and variability	• Presentation • case study videos • Plenary discussions	• Projector • Videos • Flip charts • Handouts • Laptop	20 minutes
2. Impacts of climate change and variability on agriculture and food security	• Presentation • case study videos • Plenary discussions	• Projector • Videos • Flip charts • Handouts • Laptop	15 minutes 15 minutes 15 minutes
3. Concept of Climate smart agriculture (CSA)	• Presentation • case study videos • Plenary discussions	• Projector • Videos • Flip charts • Handouts • Laptop	20 minutes 20 minutes, 20 minutes
4. Projected future climate scenarios and how to manage	• Presentation • case study videos • Plenary discussions	• Projector • Flip charts • Handouts • Laptop	15 minutes 15 minutes 15 minutes
5. Module review	• trainees' questions and comments • Facilitator' summary	• Module review	10 minutes
TOTAL			**3 hours**

1.7 Facilitators Guidelines

1.6.1 Introduction and Levelling Expectations (15 minutes)	**Session Guide**
(The facilitator welcomes trainees to the module climate change and climate smart agriculture. The facilitator then invites the trainees to introduce themselves and share their expectations on the module.	PowerPoint presentation
Module Objectives *(The facilitator levels out the trainees expectations and presents module objectives on power point)* By the end of the module the trainee should be able to: • Explain climate change and adaptations. • Define 'climate smart agriculture' • Identify impacts of climate change and variability on agriculture and food security. • Present and appreciate projected future climate scenarios and how to manage them. • Outline the benefits of climate smart crop management practices in banana production.	Distribute Participants' Handouts on Module Objectives and expectations.

1.7.2 Introduction to climate change and climate variability (45 minutes)	Session guide
The facilitator proceeds to introduce the module basics as follows: **PowerPoint presentation** • Basic terminologies used in the module (weather, climate, variability, adaptation, coping) • What is climate change and climate variability • The causes of climate change • Climate risks impacting agriculture • Proposed adaptation measures (captured in TIMPS) **Discussion** Field experiences, coping and adaptation mechanisms adopted by farmers	PowerPoint presentation Flip chart sketches Discussion
1.7.3 Concept of Climate smart agriculture (CSA) – 1 hour)	**Session Guide**
(The facilitator presents the principles underpinning CSA) **PowerPoint presentation** *(30 minutes)* • Definition of the CSA approach and their characteristics • The three pillars of CSA (productivity, Adaptation and Mitigation • Why CSA is needed **Plenary discussion***(30 minutes)*	PowerPoint presentation Handouts Plenary discussion
1.7.4 Projected future scenarios that will impact productivity (45 minutes)	**Session Guide**
(The facilitator leads the trainees in discussing future climatic projections focusing on rainfall and temperature which directly impacts on crop yields. The discussions will be as follows; **PowerPoint Presentation** • Long term rainfall and temperature projections as impacted by climate change **Video presentation** • Short Video on showing projections of rainfall and temperature **Plenary Discussion** • Projected impacts of CC on food production and required adaptation measures	PowerPoint Video presentation Plenary Discussion
1.7.5 Module Review (15minutes)	**Session Guide**
(The facilitator leads the trainees in summarizing the key points discussed in the module) Let the trainees recall the new things they have learnt in this module.	Plenary discussion

1.8 Participant's Handouts (Training and Reference Materials)

1. Esilaba, A.O.*et al*. (2019). KCEP-CRAL Climate Smart Agriculture Extension Manual. Kenya Agricultural and Livestock Research Organization, Nairobi, Kenya

MODULE 2
BANANA FARMER FIELD AND BUSSINESS SCHOOLS

2.1 Introduction to the module Farmer Field & Business School

This module is designed for exposing trainees to the basic principles of Farmer Field and Business School (FFBS) methodology. It is necessary that the FFBS facilitators have knowledge of this methodology in order to train lead farmers (LF). The LF shall facilitate other farmers enabling them to share and learn by doing, trying available technologies and innovations as they implement them on their farms and play a bigger role in marketing. In turn the LF shall facilitate farmer led FFBS.

The traditional extension methods involve technology and advisory services dispensing technologies to farmers whose inputs arenot soughtand who are not adequately involve in deciding what they want. The methodologies do not give space for farmers' involvement in identifying problems, evaluating and implementing possible solutions. The methods have been successful to some extent but are inadequate in addressing some needs including market needs.

There is therefore need for alternative sustainable approaches that will assist farmers to face and adopt new technologies and innovations in a changing banana farming environment. The approach builds on the successes and lessons learnt from Farmer Field School (FFS)which emphasizes improved field productivity without considering the economic, socio-cultural and political frameworks around which the FFS are run. However, the FFBS approach lay emphasis on community approaches that plan to facilitate market engagement, nutrition, gender and performance monitoring and evaluation.

The FFBS is a capacity building method based on adult education principles through learning by doing. It is dynamic and involves interaction between farmers and stakeholders thereby building farmers' capacity to identify production and marketing constraints and tests possible solutions through learning by doing. It also focuses on farming as a business, building farmers' capacity as entrepreneurs and managers' and empowers them with leadership, communication and business skills. Being participatory, FFBS improves the learners' observation skills and creates linkages with other value chain players. Lead farmers graduating from FFBS become experts to start other farmer led FFBS reaching more farmers sustainably.

2.2. Module Learning Outcomes

By the end of the module the following should be achieved:

- The success and limitations of existing extension approaches identified.
- The concept of FFBS as a 'learning by doing' approach described and appreciated.
- Logical sequences of events for a successful FFBS program developed

- The FFBS methodology in addressing the limitations of existing extension approaches applied.

2.3 Module Target Group

This module targets public agricultural extension and private service providers based at county, sub-county and ward level. It will also be useful for private extension service providers dealing directly with farmer groups at community level.

2.4 Module users

This module is intede for use by master trainers who are members of the core team of trainers (CTT) and lead farmers in the bananavalue chain target counties. The trainers using this module should thoroughly farmilialize themselves with participants handouts (Training materials)

2.5 Module Duration

The Module is estimated to take a minimum of 4 hours 40 minutes

2.6 Module Summary

Farmer Field and Business School Approaches			
Sessions	**Training Methods**	**Training Materials**	**Time**
2.6.1 Introduction and Leveling of expectations	• Buzz • Presentation	• Module objectives	15 minutes
2.6.2 Existing agricultural extension approaches	• Brainstorming • Presentation	• Flip charts, felt pens • Case study, flash cards • Flip charts, felt pens	30 minutes 20 minutes
2.6.3 Facilitation approaches	• Brain storming • Presentation • Discussion	• Flip charts, felt pens • PowerPoint slides	20 minutes 20 minutes 20 minutes
2.6.4 Introduction to FFBS	• Presentation • Plenary discussion on experiences on FFS • Group exercise • Presentation	• PowerPoint • Flip chart, participants handouts • Exercise guide	10 minutes 30 minutes 30 minutes 10 minutes
2.6.5. Key steps in an FFBS establishment and capacity building model	• Presentations and discussions • Group work presentation	• Projector • Participants handouts	30 minutes 30 minutes
2.6.6. Review of Module	• Trainees' questions and comments • Facilitator's summary	• Participants handouts	15 minutes
TOTAL			**4 hours 40 minutes**

2.7.1. Introduction and Levelling of Expectations (15minutes)	Session Guide
(The facilitator welcomes trainees to the module and invites them to state their expectations on the module). The facilitator then levels off the trainees' expectations and states the module objectives. **PowerPoint presentation** **Module Objectives** The facilitator presents module objectives. By the end of the module the trainee should be able to: • Identify success and limitations of existing conventional extension approaches. • Describe the concept, characteristics, principles and plans of Farmer Field and Business School (FFBS) as a 'learning by doing' approach. • Develop a logical sequence of events for a successful FFBS program . • Apply FFBS methodology in addressing the limitation of existing extension approaches.	• List the trainees' expectations on a flip chart and clearly display them for all to see. • PowerPoint presentation displayed. • Distribute participants handout

2.7.2 Existing Agricultural Extension (40 Minutes)	Session Guide
(Facilitator guides trainees in defining agricultural extension and discussing existing agricultural extension approaches and the characteristics, success and limitations of traditional extension methods).	Agree on the definition
PowerPoint presentation • Define Agricultural extension • What are the existing agricultural extension approaches? • Definitions and Approach. The facilitator then asks trainees to state the various types of extension approaches that they know of. **PowerPoint presentation/ Group Discussion** Categories of extension approaches. • Present/Discuss technology transfer extension approach and challenges. • Present/Discuss Advisory extension approach • Present/Discuss Facilitation extension approach.	List the approaches on a flip chart. Distribute participants handout Presentation on slides or flip charts

2.8 Participant's Handouts (Training and Reference Materials)

1. FAO GOVERNMENT COOPERATIVE PROGRAMME: FARMER FIELD AND FARM BUSINESS SCHOOLS. Manual for Preparation and Establishment of Farmer Field and Farm Business Schools
2. Khisa Godrick: Farmer Field School Methodology, Training of Trainers Manual.
3. FAO: Farmer Field School Methodology, a TOT Manual
4. Sustainet East Africa: Farmer Field School, A Technical Manual
5. Training Manual on Good Agricultural Practices in Potato using Farmer Field and Business School Approach: KALRO AND GiZ initiative

MODULE 3

BANANA PRODUCTION NICHE AND CLIMATIC REQUIREMENTS

3.1 Introduction

This module exposes service providers and lead farmers to the different types of ecological conditions (altitudes, soils, AEZs and climatic conditions) suitable for banana production in the selected counties. Banana is grown from 0 to 1800 meters above sea level under rain-fed conditions primarily by small-scale farmers. The average banana holding in Kenya is 0.3 hectares, with nearly every rural household having at least a couple of banana mats around the homestead. Although production in terms of area planted and quantity produced has increased over the years, yield per hectare has remained stagnant ranging from 4.5 to 10 tons per hectare compared to a potential of 30 to 40 tons per hectare. For a crop to realize its full genetic potential, it must be planted in the environment that suits it most.

3.2 Module Learning outcomes

By the end of this module training the following should be achieved:

- Importance of Banana in Kenya's economy appreciated.
- Altitudes and soil types/characteristics for banana production mapped and shared.
- Climatic conditions (temperatures, rainfall and humidity) required for Banana production identified and shared.
- County agro-ecological zones for Banana production discussed.

3.3 Module Target Group and Categories

This module is intended for public and private agricultural extension service providers in the banana value chain target counties.

3.4 Module users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT) and Lead Farmers in the banana value chain target counties. The trainees' using this module should thoroughly familiarize themselves with the participant's handouts (training materials).

3.5 Module Duration

The module is estimated to take a minimum of 4 hours

3.6 Module Summary

Module 3: Banana production niches and climatic requirements			
Sessions	**Training methods**	**Training materials**	**Time**
3.6.1 Introductions and climate setting	• Presenter introductions • Self-introduction of trainees (incl. individual involvement in bananas • Plenary discussions	• Flips charts • Felt pens • Projector • Laptop	30 minutes
3.6.2 Objectives and expectations	• Presentations • County group exercise (trainees list expectations) • Plenary discussions to share expectations	• Flips charts • Felt pens • Laptop for power point Presentation	1 hour
3.6.3 Importance of banana in Kenya's economy	• Presentations • Plenary discussions	• Flips charts • Felt pens • Projector • Handouts • Laptop	15 minutes
3.6.4 Banana production ecological/climatic requirements for optimal yields	• Presentations • Plenary discussions	• Flips charts • Felt pens • Laptop • Projector • Handouts (Training notes)	30 minutes
3.6.5 Banana production Agro-ecological zones (AEZs)- average yields, and constraints in the target Counties	• Group work identify banana production pockets in their sub-Counties/Counties • Presentations • Plenary discussions • Field demonstration	• Flips charts • Felt pens • Projector • Laptop	1 hour 30 minutes
3.6.6 Module review	• Discussions/ conclusion and way forward	• Flip charts • Projector • Laptop	15 minutes
Total			**4 hours**

3.7 Facilitator's Guidelines

Module 3: Banana production and appropriate climatic requirements	
3.7.1. Introductions and climate setting (30 minutes)	**Session Guide**
The facilitator welcomes trainees to the module and invites to introduce themselves and state their expectations on the module	• Summarize the facilitator/ trainees involvement in bananas value chains
3.7.2. Objectives and expectations (1 hour)	
(Facilitator then states the module objectives and levels off the trainees expectations) **PowerPoint presentation** **Objectives (15 minutes)** By the end of this training module the trainee should be able to: • Appreciate the importance of banana in Kenya's economy. • Describe soil types/characteristics for bananas production. • Describe and identify climatic conditions (temperatures, rainfall and humidity) required for banana production. • Identify the banana suitable agro ecological zones within their counties. **Group exercise (45 minutes)** The trainees break into groups (e.g. county based) and list their expectations from the module.	• PowerPoint presentations • Group exercise (listing and presenting expectations). • Expectations lists kept for later reviewing compliancy

3.7.3 Importance of banana in Kenya's economy (15 minutes)	
The facilitator presents and guides in group plenary discussion. **PowerPoint Presentation (10 minutes)** • Origin and place of bananas as crop. • Why bananas in Kenyan households • Key counties producing bananas in Kenya. • General banana production in Kenya. **Facilitator's guided discussions (5 minutes)** Questions/answers/comments	• PowerPoint Presentation • Distribute to participants Handouts (training materials)
3.7.4 Banana production ecological/climatic requirements (30 minutes)	
Facilitator presents and then guides a discussion on importance of banana in Kenya's economy **Plenary Presentation (20 minutes)** Importance of banana in Kenya's economy • Altitude and Agro-ecological zones. • Climatic conditions (Rainfall, Temperatures and humidity). • Soils (soil types, pH, general fertility for bananas). **Facilitator's guided discussions (10 minutes)** Questions/answers/comments	• PowerPoint Presentation • Distribute to participants Handouts (training materials)

3.7.5. Banana production AEZs (villages), average yields, and constraints in the target Counties (1 hour)	Session Guide
Facilitator presents and then divides trainees in to groups ready for group work. **Plenary Presentation/Discussion (30 Minutes)** Facilitator guides in reviewing and discussing suitability map (county by county). **Group work (30 minutes)** Trainees to bring out specific County or sub-county AEZs, land size, yields and constraints to banana production. Then, the trainees present in the plenary: • Agro-ecological zones (AEZs) and % area suitable for bananas • Average land/farm size under bananas. • Average banana farm yields. • Constraints to bananas production. **Discussions/presentations from the groups (30 minutes)** Let the trainees/groups share the exercise outcomes	• PowerPoint presentations • Group work • Open but facilitator's guided discussions
3.7.6. Module review (1 hour)	**Session Guide**
(The facilitator leads the trainees in reviewing the module). Summary of the main points from the training (45 minutes). • Objectives and expectations (review done on basis of the earlier listed objectives and expectations). • Bananas production ecological/climatic requirements Banana production AEZs (villages) average yields, and constraints in the target counties. • Randomly (average of 10 cases), *trainees indicate new thing(s) learned from the module. The results are recorded per county presented.* • Randomly (average of 10 cases) trainees pin-point the way forward issues. **Facilitator's guided discussions (15 minutes)** • Facilitator summarizes the main points of the module on a flip chart and displays.	• The last Participants' Handouts/training materials • Summary of the main points

3.8 Participants' Handouts and reference

1. Banana production Technical Guide [Manual, 2015]
2. Banana leaflets [2017]

MODULE 4
IMPROVED BANANA VARIETIES

4.1 Introduction to the Module

This module is intended to expose trainees to the available improved Banana varieties. The trainees shall be empowered to facilitate farmers, enabling them share and learn by doing, trying available technologies and innovations as they implement them on their farms. It also seeks to promote resilience among banana farmer through improved productivity, climate adopted varieties, better fruit quality and yield.

4.2 Module learning out comes

The local banana varieties are of low quality and low economic value thus the need for banana farmers to adopt improved varieties which will enable them to get higher returns. Besides taking too long to come into production, most of the local varieties are too tall for effective management and fruits are small and fibrous thus not preferred in the market. On the other hand, the improved varieties take shorter time to bear, have bigger fibreless fruits which are greatly preferred in the market and fetch higher returns. In addition, they are easier to manage due to their characteristic short height and tolerance to disease. Improved varieties are also preferred by exporters and processors. Processors prefer improved varieties due to greater recovery rate and their fibreless nature which make them easier to process.

By the end of the module trainees should achieve the following:

- Successes and limitations of existing local varieties are identified.
- Characteristics of the improved varieties are described.
- Use of improved varieties in addressing the limitation of existing local varieties appreciated.
- Plan for promotion of successful adoption of improved varieties under FFBS program developed and shared.

4.3 Module Target group

This module targets public and private agricultural extension service providers based at sub county and ward levels.

4.4 Module users

This module outlines the learning outcomes, the category of trainees targeted, module summary, facilitator's guidelines and participants' handouts. The trainers using this module should thoroughly familiarize themselves with the participants' handouts.

This module can be used by master trainers and members of CCTS who are members of the Core Team Trainers.

4.5 Module duration

The Module is estimated to take a minimum of 4 hours and 20 minutes

4.6 Module Summary

Improved Banana varieties

Sessions	Training Methods	Training Materials	Time
4.6.1.Leveling of expectations	• Buzz • Presentation	• Module objectives	15 minutes
4.6.2 Existing/ local Banana varieties	• Brainstorming • Presentation	• Flip charts, felt pens • Case study, flash cards • Flip charts, felt pens	20 minutes 20 minutes
4.6.3.Limitation of local varieties	• Brain storming • Presentation • Discussion	• Flip charts, felt pens • Projector • Laptop	20 minutes 15 minutes 10 minutes
4.6.4. Introduction to improved Banana varieties, characteristics	• Presentation • Plenary discussion on experiences on FFS • Group exercise • Presentation	• Projector • Flip chart, participants handouts • Exercise guide • Laptop	
4.6.5. Uses of Improved Banana varieties	• Presentations and discussions • Group work presentation	• Projector • Participants handouts • laptop	20 minutes 45 minutes
4.6.6.Review of Module	• Participants' questions and comments • Facilitator's summary	• Participants handouts	15 minutes
TOTAL			**4 hours 20 minutes**

4.7 Facilitator's Guidelines

<table>
<tr><th colspan="2">Improved Banana Varieties</th></tr>
<tr><td>4.7.1 Introduction, objectives and expectations (15 minutes)</td><td>Session Guide</td></tr>
<tr><td>Introduction
(The facilitator welcomes trainees to the module
The facilitator invites the trainees to introduce themselves and state their expectations).

Module Objectives
(The facilitator presents the objectives of the module)
By the end of the module the trainee should be able to:
• Identify successes and limitations of existing local varieties.
• Describe characteristics of the improved varieties.
• Appreciate use of improved varieties in addressing the limitation of existing local varieties.
• Develop and share plan for promotion of successful adoption of improved varieties under FFBS program.</td><td>Summarize trainees "expectations" and display

Distribute handouts to participants
• Module objectives
• Program</td></tr>
<tr><td>4.7.2 Existing/local Banana varieties (40 minutes)</td><td></td></tr>
<tr><td>The facilitator should engage the trainees in identifying the existing local varieties their characteristics, success and limitations.

PowerPoint Presentation
• Existing varieties
• Characteristics of local varieties
• Success and limitation of local/existing varieties</td><td></td></tr>
<tr><td>4.7.3 Introduction to Improved Banana varieties, Characterization and ecological requirements (45 minutes)</td><td>Session guide</td></tr>
<tr><td>(The facilitator should identify and describe the characteristics of improved varieties and their ecological requirements)

Plenary Presentation
Facilitator presents improved banana varieties and describes the characteristics of improved varieties based on
• Disease tolerance
• Drought tolerance
• Early maturity
• High yielding
• Taste, and market demand
Plenary Presentation/Discussion
Describe the Ecological requirements of improved banana varieties, give examples of suitable locations where various varieties can do well. Explain how the following Biotic conditions influence production
• Rainfall, Soil,Temperature and Humidity</td><td>Distribute handouts to participants

List of banana varieties and their characteristics and suitable ecological zones</td></tr>
</table>

<table>
<tr><th>4.7.4 Uses of improved Banana varieties (1 hour 20 minutes)</th><th>Session guide</th></tr>
<tr><td>Facilitator presents the following on power point slides and flip charts:

Plenary Presentation
To identify uses of improved banana varieties.
Cooking type
Desert
Apple
Dual purpose</td><td>Distribute handouts to participants
Improved Banana varieties and their uses
Q & A sessions</td></tr>
<tr><td>4.7.5 Practical aspects of improved banana varieties in the orchard (1 hour 50 minutes)</td><td>Session guide</td></tr>
<tr><td>Facilitattor leads trainees to the field for practical demonstration on how (1 hours):
Group Work
To identify different varieties
To distinguish between cooking and ripening varieties
Maturity indices
Discussion (50 minutes)
After the group work allow trainees to raise any issues and discuss them.</td><td>Distribute handouts to participants
Manuals
Brochures
Posters of different Banana varieties
Leaflets</td></tr>
<tr><td>4.7.6 Model review (15 minutes)</td><td>Session guide</td></tr>
<tr><td>(The facilitator leads the trainees in reviewing the module)

Summarize the main points of the training
Together with the trainees review the main points in improved banana varieties.
What new things did you learn from this module?
What are some of the problems and issues that you have become more aware of in the module?</td><td>Group discussions

Q & A session
Recap the main points
Test understanding
Link to the way forward
Participatory evaluation of the session</td></tr>
</table>

4.8 Participant's Handouts (Training and Reference Materials)

1. Banana production Technical Guide [Manual, 2015]
2. Banana leaflets [2017]

MODULE 5
SEED SYSTEM-BANANA PROPAGATION AND HARDENING NURSERIES

5.1 Introduction to the Module

This module is intended to expose trainees to the protocols to be used in propagation of selected popular banana varieties in Kenya namely: Ngombe, Nusu Ngombe, Grand Naine, Chinese Cavendish, FHIA 17 and Williams hybrid. The trainees shall be empowered to facilitate farmers, learning the banana propagation techniques. It also seeks to improve the availability of clean planting materials by promoting among banana farmers, the adoption of new knowledge and skills for producing quality planting material in order to meet the high demand for clean banana planting material in Kenya for improved banana productivity.

5.2 Module Learning Outcomes

By the end of the module, the trainee should achieve the following:

- Trainees equipped with skills on banana propagation techniques.
- Appropriate propagation techniques in banana production learnt and applied
- Adequate skills on banana propagation for increased productivity and yields provided.
- Banana propagation in hardening nursery enhanced.

5.3 Module Target Group

This module targets all agricultural extension officers, banana technicians and all private banana planting material producers countrywide.

5.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The trainers using this module should thoroughly familiarize themselves with the participants handouts.

5.5 Module Duration

The Module is estimated to take a minimum of 4 hours 30 minutes

5.6 Module Summary

Propagation of clean banana planting materials			
Sessions	**Training Methods**	**Training Materials**	**Time**
5.6.1 Introduction, objectives and expectations	• Personal introductions • Presentations	• Projector • Flip Chart • Felt pens	15 minutes

5.6.2 Banana sucker selection, uprooting and paring (Macro propagation techniques)	• Presentations • Plenary discussions • On-farm demonstrations	• Flips charts • Felt pens • Projector • Handouts	1 hour
5.6.3 Tissue culture, banana plantlet hardening and macro propagation techniques	• Presentations • Plenary discussions	• Flips charts • Felt pens • Projector • Handouts	30 minutes
5.6.4 Practical aspects of banana sucker uprooting, paring, hot water treatment, media sterilization, macro propagation, potting, hardening	• Plenary discussions • Practical demonstrations in clean banana propagation techniques and nursery management	• Flips charts • Felt pens • Projector • Handouts	2 hours 30 minutes
5.6.5 Module review	• Presentations • Plenary discussions	• flips charts • Projector • Handouts	15 minutes
TOTAL			**4 Hours 30 minutes**

5.7 Facilitator's Guidelines

Banana Propagation and Hardening Nurseries	
5.7.1 Introduction, objectives and expectations (15 minutes)	**Session Guide**
Introduction *(The facilitator welcomes trainees to the module to introduce themselves by stating their profile and experience* *The facilitator invites the trainees to introduce themselves and state their expectations).* **PowerPoint Presentation** Module Objectives (*The facilitator presents the objectives of the module*) By the end of the module trainee should be able to: • Equip leaners with skills on banana propagation techniques. • Apply appropriate propagation techniques in banana production. • Provide adequate skills on banana propagation for increased productivity and yields. • Enhance banana propagation skills in hardening nursery.	Summarize trainees "expectations" and display Distribute handouts to participants • Module objectives • Program

5.7.2 Banana macro propagation techniques (1 hour)	Session guide
(*The facilitator should describes innovative banana propagation techniques*). **PowerPoint Presentation (30 Minutes)** • Application of appropriate propagation techniques in banana sucker selection, and paring. • Clean starter material obtained through paring of suckers; hot water treatment of suckers; **Group discussion (30 minutes)** • Banana macro propagation techniques	Distribute handouts to trainees Macro propagation techniques brochures
5.7.3 Banana plantlet hardening techniques (30 minutes)	**Session guide**
(The facilitator guides the trainees in identifying challenges in acquiring clean banana planting materials and how to address them by establishing hardening nurseries and macro propagation units). **PowerPoint Presentation (20 Minutes)** • Description and Importance of a hardening nursery in raising Banana seedlings. • Types of Hardening Nurseries; Materials required for construction of a hardening nursery. • Site selection for a Banana Hardening Nursery, (proximity to a water source, accessibility and security). • Tools and equipment needed in a hardening Nursery. • Sourcing for clean planting material for hardening nurseries (Tissue Culture and macro propagation). • Management practices in the Hardening Nursery. **Group discussion** Facilitator asks trainees to name common sources of planting material used by farmers and their challenges. Focus discussion on: • Sources of banana planting materials. • Challenges' encountered by farmers when sourcing for banana planting material.	Distribute handouts to participants • manuals, poster, brochures and leaflets Q & A sessions
5.7.4 Practical aspects in Macro propagation and Hardening nurseries (2 hours 30 minutes)	**Session guide**

<table>
<tr><td>Facilitator the trainees proceed to the field for practical demonstration on;
Banana sucker selection and uprooting
Pairing
Hot water treatment
Media sterilization
Macro propagation
Potting
Hardening</td><td>Distribute handouts to trainers
Posters</td></tr>
<tr><td>5.7.5 Model review (15 minutes)</td><td>Session guide</td></tr>
<tr><td>(The facilitator should leads the trainees in reviewing the module)

Facilitator summarize sthe main points of the training
Together with the trainees review the main points in banana Macro propagation and Hardening Nurseries
What new things did you learn from this module?
What are some of the problems and issues that you have become more aware of in the module?</td><td>Group discussions

Q & A session
Recap the main points
Test understanding
Link to the way forward
Participatory evaluation of the session</td></tr>
</table>

5.8 Participants' Handouts (Training and Reference Materials)

1. IITA-Training Manual on macro-propagation of banana and plantain
2. Procured training materials for farmer groups

MODULE 6

BANANA AGRONOMY (ORCHARD ESTABLISHMENT AND MANAGEMENT)

This module outlines the learning outcomes, the category of trainees targeted, module duration, users and summary, facilitator's guidelines and participants' handouts.

6.1 Introduction to the Module

This module exposes the trainees to the protocols to be used in banana orchard establishment and management for enhanced productivity. It empowers them with appropriate knowledge on banana management practices which include land preparation, field lay out, hole digging, manure and fertilizer application, planting, de-suckering, pruning, propping, basin formation, weeding and water management. Trainees will be able to impart this knowledge to farmers thus enabling them to enhance the production of quality bananas, to meet the high demand for quality bananas in Kenya.

6.2 Module Learning Outcomes

By the end of the training module, the following should be achieved:

- Trainees equipped with skills on banana orchard establishment and management.
- Appropriate agronomic management practices to increase productivity and quality identified and shared.
- Adequate skills on use of various inputs to optimize productivity shared.
- To reduce biotic and abiotic stresses.
- Knowledge on orchard establishment shared and tried.

6.3 Module Target Group

This module targets all extension officers, agriculture technicians and all private service providers and lead farmers.

6.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The trainers using this module should thoroughly familiarize themselves with the participant's handouts.

6.5 Module Duration

The Module is estimated to take a minimum of 5 hours 30 minutes.

6.6 Module Summary

Banana orchard establishment and management			
Sessions	**Training Methods**	**Training Materials**	**Time**
6.6.1 Introduction, objectives and expectations	• Personal introductions • Presentations	▪ Projector	15 minutes
6.6.2 Site selection, land preparation, field lay out, hole digging	• Presentations • Plenary discussions • On-farm demonstrations	• Flips charts • Felt pens • Projector • Handouts	1 hour 30 minutes
6.6.3 manure and fertilizer mixing, planting, de-suckering and pruning	• Presentations • Plenary discussions	• Flips charts • Felt pens • Projector • Handouts	30 minutes
6.6.4 Basin formation weeding, mulching, watering	• Presentations • Plenary discussions	• Flips charts • Felt pens • Projector • Handouts	30 minutes
6.6.4 Practical aspects on planting, fertilizer (organic and inorganic) application, planting de-suckering, weed control, basin formation, irrigation, (watering), pruning and propping	• Presentations • Plenary discussions • Practical demonstrations in orchard establishment management	• Flips charts • Felt pens • Projector • Handouts	2 hours 30 minutes
6.6.5 Model review	• Presentations • Plenary discussions	• flips charts • Projector • Handouts	15 minutes
TOTAL			**5 hours 30 minutes**

6.7 Facilitator's Guidelines

Banana orchard establishment and maintenance	
6.7.1 Introduction, objectives and expectations (15 minutes)	**Session Guide**
Introduction *(The facilitator introduces the module and invites trainees' to introduce themselves and state their expectations. The facilitator then presents module learning outcomes and expectations).* **Module Objectives** (*The facilitator presents the objectives of the module*) By the end of the module trainee should be able to: • Equip the learner with skills on banana orchard establishment and management • Apply appropriate agronomic management practices to increase productivity and quality • Provide the learner with adequate skills on how to use various inputs to optimize productivity. • Reduce biotic and abiotic stresses.	Summarize participants "expectations" and display Distribute handouts to participants • Module objectives • Program
6.7.2 Site selection, land preparation, field lay out, hole digging (1 hour 30 minutes)	**Session guide**
(*The facilitator makes powerpoint presentation on the following*). • Selecting suitable site • Clearing the site if bushy • Removal of perennial weeds using herbicide if necessary • Field lay out • Hole digging **Discussion (30 minutes)**	Distribute handouts to participants on • Site selection and • preparation • Laying out • Hole digging
6.7.3 Banana planting, Nutrient management and de-suckering (30 minutes)	**Session guide**
(Facilitator presents the following on power point slides and flip charts): • Different source of fertilizers (organic and inorganic) • Mixing of top soil with manure and phosphatic fertilizer • Planting and firming the soil around the seedling • Watering requirement, use of Zai pit for water harvesting • Procedures for de-suckering and pruning. **Discussion (30 minutes)** After the presentation, facilitator guides the trainees in recalling what they learned and discussing issues that may arise.	Distribute handouts to participants • Planting • Agronomic requirements • Orchard management Q & A sessions

<table>
<tr><td>6.7.4 Basin formation weeding, mulching, watering (30 minutes)</td><td>Session guide</td></tr>
<tr><td>(Facilitator presents the following on power point slides and flip charts):
• Basin formation
• Weed control methods
• Mulching material and application
• Watering method and frequency
• Intercropping-
Discussion (30 minutes)
After the presentation, the facilitator guides the trainees in recalling what they learned and discucing issues that may arise.</td><td>Distribute handouts to participants
• Orchard Management

Q & A sessions</td></tr>
<tr><td>6.7.5 Practical aspects of banana orchard establishment and management (2 hours 30 minutes)</td><td>Session guide</td></tr>
<tr><td>(Fcalitator leads trainees to the filed for demonstration on the following) (2 hours 30 minutes):
• How to select suitable site
• Lay out, spacing
• Hole digging
• How to apply fertilizer (organic and inorganic), rate and placement
• How to plant
• How to make basins, and weeding
• De-suckering, pruning and propping
• Zai pit- how to make them
• Irrigation-(drip)
Discussion (30 minutes)
• After the presentations the facilitator allows trainees to raise any issues and discuss them.</td><td>Distribute handouts to participants
• Posters
• Leaflets</td></tr>
<tr><td>6.7.6 Model review (15 minutes)</td><td>Session guide</td></tr>
<tr><td>(The facilitator leads the trainees in reviewing the module).

Summarize the main points of the training
• Together with the trainees review the main points in banana orchard establishment and management
• What new things did you learn from this module?
• What are some of the problems and issues that you have become more aware of in the module?</td><td>Group discussions

Q & A session

• Recap the main points
• Test understanding
• Link to the way forward
• Participatory evaluation of the session</td></tr>
</table>

6.8 Participant's Handouts (Training and Reference Materials)

1. Banana Training Manuals
2. Banana brochure
3. Procured training materials for farmer groups

MODULE 7

INTEGRATED SOIL AND WATER MANAGEMENT PRACTICES FOR BANANA PRODUCTION

7.1 Introduction to the module

The performance of the agriculture sector in Kenya has continued to decline over the years due to increased soil acidity, mining of nutrients not supplied in the applied fertilizers and lowering of the soil organic matter content caused by non-use of organic resources. Macronutrients (nitrogen (N), phosphorus (P), potassium (K) and Sulphur (S) and micronutrients (zinc (Zn), Molybdenum (Mo) and Boron (B) have been identified as deficient in Kenyan soils. Additionally, climate change has accelerated the decline of the agricultural sector performance through limited and unpredictable water supply to crop production systems. Integrated Soil Fertility Management (ISFM) which includes conservation agriculture offers the best options for improving soil fertility while allowing for climate change adaptation.

Banana is mostly cultivated by smallholder farmers with minimal inputs. Drought management technologies to mitigate drought effects in the banana production are available. However, farmers have not realized the full benefits due to limited integration of the developed Integrated Natural Resource Management (INRM) and sustainable intensification practices in their banana production systems.

This module introduces service providers, lead farmers and trainees to the importance of integrated soil and water management practices for enhanced banana production.

7.2 Module learning outcomes

By the end of the module, the following training outcomes should be achieved:

- Knowledge on soil composition, the various physical, chemical and biological properties and what constitutes a healthy soil including soil classification acquired.
- Soil and plant tissue sampling for laboratory analysis, interpretation and utilization of results and also identification of accredited laboratories in Kenya discussed and shared.
- Soil fertility and plant nutrition for increased crop productivity (R4 Stewardship that includes fertilizer source, rates, application methods and timing) appreciated
- Knowledge on soil health and Integrated Soil Fertility Management (ISFM) for climate resilient cropping systems (conservation agriculture, cereal legume

intercrops, rotation, organic amendments, manure and composting) acquired.
- Knowledge on water harvesting technologies, soil and water management acquired.
- Knowledge and skills for identifying temporary or permanent decline of land productive capacity and provide various solutions to soil degradation imparted and appreciated.
- Awareness on the occurrence of problematic soils and their management increased appreciated.

7.3 Module Target Group

This module is intended for service providers, and county public extension agents and lead farmers in the banana producing areas.

7.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The trainers' using this module should thoroughly familiarize themselves with the participants handouts.

7.5 Module Duration

The Module is estimated to take a minimum of 6 hours

7.6 Module Summary

Module 7: Integrated soil and water management practices for banana production			
Sessions	**Training methods**	**Training materials**	**Duration**
7.6.1 Introduction, objectives and expectations	• Personal introductions • Presentations • Plenary discussions	• Flips charts • Projector • Laptop	15 minutes
7.6.2 Soil composition, properties and health **7.6.2.1** Conduct Soil sampling and analysis practical's	Presentations Field demonstrations	• Flips charts • Projector • Participants • Handouts • Laptop	45 minutes

7.6.3 Soil and plant tissue sampling and analysis	Presentations Field demonstrations Field demonstrations	• Flips charts • Projector • Laptop • Participants Handouts Sampling and analysis tools, equipment and materials	1 hour 15 minutes
7.6.4. Soil fertility and plant nutrition	Presentations Field demonstrations	• Flips charts • Projector • Laptop • Participants Handouts	30 minutes
7.6.5 Soil health and ISFM for climate resilient cropping systems	Presentations Field demonstrations	• Flips charts • Projector • Participants • Handouts	45 minutes
7.6.6 Soil, water management and water harvesting technologies	Presentations Field demonstrations	• Flips charts • Projector • Laptop • Participants Handouts	1 hour
7.6.7 Soil degradation and reclamation	Presentations Field demonstrations	• Flips charts • Laptop • Projector • Participants Handouts	30 hours
7.6.8 Problematic soils and their management	Presentations Field demonstrations	• Flips charts • Projector • Participants Handouts • Laptop	30 minutes

7.6.9 Module review and discussion	Discussions	• Flips charts	30 minutes
Total			**6 hours**

7.7 Facilitator's Guidelines

Module 7: Integrated soil and water management practices for banana production

7.7.1. Introduction, Objectives and Expectations (15 minutes)	Session Guide
(The facilitator introduces the module and invites trainees to introduce themselves by stating their profile and experience and state their expectations. The facilitator then presents modules learning outcomes and expectations) **Module Objectives (15 minutes)** *(The facilitator presents module objectives)* By the end of the module the trainee should be able to: • Appreciate soil composition, properties and health. • Explain how soil and plant tissue sampling and analysis is done. • Relate soil fertility to plant nutrition. • Identify Soil health and integrated soil fertility management) for climate resilient cropping systems. • Acquire essential soil and water management and water harvesting technologies. • Acquire knowledge for soil degradation and reclamation. • Appreciate problematic soils and their management .	• Summarize trainees' "Expectations" and display. • PowerPoint Presentation • Distribute Participants Handouts on Module Objectives and Training Program

7.7.2. Soil composition, properties and health (45 minutes)	Session Guide
(The facilitator presents on soil composition, properties and health) **Plenary Presentation (30 minutes)** • Soil composition, properties and health • Description of soil composition • Description of soil properties • Describe what soil health is all about **Discussion (15 minutes)** Facilitator allows the trainees to recall what they learned and discuss any issue that may arise	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual

7.7.3. Soil and plant tissue sampling and analysis (1 hour)	Session Guide
Plenary Presentation (15 minutes) • Overview of the soil sampling methods • Soil analysis results and interpretation • Overview of soil analysis results using available examples • Soil sampling guidelines **Practical exercise on soil sampling (1 hour 15 minutes)** Demonstration on soil sampling method	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manuals Practical Demonstration

7.7.4. Soil fertility and plant nutrition (30 minutes)	Session Guide
Plenary Presentation(15 minutes) • Potential role of different soil managements techniques in addressing soil fertility challenges in banana smallholder farming systems • Integrated Soil Fertility Management techniques • Soil management guidelines **Discussion (15 Minutes)** The facilitator allows the trainees to recall what they learned and discuss any issue that may arise.	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual

7.7.5 Soil health and (ISFM) for climate resilient cropping systems (45 minutes)	Session Guide
Plenary Presentation (30 minutes) • Soil health • Integrated soil fertility management (ISFM) • Relate soil health and ISFM for a climate resilient cropping system • Manure management, mulching, organic amendments and composting for increased use of organic manure for improving agricultural production • Conservation agriculture as a climate smart agriculture practice • Relate banana legume intercrops and crop rotation as climate resilient cropping systems **Discussion (15 Minutes)** Facilitator allows the trainees to recall what they learned and discuss any issue that may arise.	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manuals

7.7.6 Soil and water management and water harvesting technologies (1 hours)	Session Guide
Plenary Presentation (45 minutes) • Principles of soil management for increased crop productivity • Methods of tillage systems that conserve water for crop use. • Methods of soil fertility management for increased crop productivity **Discussion (15 Minutes)** Facilitator allows the trainees to recall what they learned and discuss any issue that may arise.	**Session Guide** • PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual
1.1.7 Soil degradation and reclamation (30 minutes)	**Session Guide**
Plenary Presentation (15 minutes) • Overview of soil degradation and reclamation. • Causes of soil degradation • Identification of Reclamation measures of degraded soil **Discussion (15 Minutes)** Facilitator allows the trainees to recall what they learned and discuss any issue that may arise.	• **Session Guide** • PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual
7.7.8 Problematic soils and their management (30 minutes)	**Session Guide**
Plenary Presentation (15 minutes) • Problematic soils and their management • Soils with unsuitable biological properties • Soils with unsuitable chemical properties • Soils with unsuitable physical properties **Discussion (15 Minutes)** Facilitator allows the trainees to recall what they learned and discuss any issue that may arise.	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual

7.7.9. Module review (30 minutes)	Session Guide
The facilitator leads the trainees in reviewing the module) Summarize the main points of the training module and together with the trainees review the main points. *(Discuss with trainees the new things learnt from this Module. What are some of the problems and issues that they have become more aware of in the module?)*	· The last Participants' Handouts · Summary of the main points on from the module on a flip chart and display

7.8 Participants' Handouts and reference materials

1. Banana Extension Manual [KCEP-CRAL Manual, 2019]
2. Banana Leaflets [KCEP-CRAL Manual, 2019]
3. Soil Management Extension Manual [KCEP-CRAL Manual2019]
4. Soil Management Leaflets [KCEP-CRAL PAMHPLETS2019}

MODULE 8

CROP HEALTH

8.1 Introduction to the module

Banana is an important crop in Kenya with a high market demand for quality fruits and its value added products. However, banana production faces serious challenges due to numerous pests that cause reduced yield, quality, and increased cost of production. Pests such as weevils, nematodes and fusarium wilt, survive for long periods in soils once established and are a major challenge in banana production. This is especially so due the fact that banana is a perennial crop. Arthropod pests such as thrips cause surface damage to the fruit, mostly reducing marketability. Banana weevils attack the rhizome and pseudo stem causing yield reductions through plant death, delayed maturation, reduced bunch weight and create avenues for pathogen entry. To ensure sustainable production of high quality banana fruits, there is need to apply effective strategies to control pests. These include tolerant cultivars, biological control, cultural practices, exclusion and judicious use of pesticides. This module is intended to expose trainees to the major pests, diseases and weeds constraining banana production, their damage, impact, and control measures.

8.2 Module learning outcomes

By the end of this module the following should be achieved:

- Major banana pests of economic importance including emerging and migratory ones identified.
- Sustainable integrated banana pest management practices and scouting for threshold determination shared.
- Major banana diseases and nematodes that cause economic losses, conditions that favour their development and their management strategies identified and shared.
- Sustainable integrated banana diseases and nematode management, scouting for threshold determination and control discussed.
- Propagation of pest/disease free planting materials identified.
- Alternative management options for pest management identified.
- Major weeds of economic importance in banana production systems and their management discussed.
- Safe and effective use of pesticides (application rate/ calibration, protective gear, pre-harvest interval) and disposal of residual/ containers; use of up to date source of registered pesticides discussed and shared.

8.3 Module Target Group

This module is intended for service providers, county public extension agents and lead farmers.

8.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The trainers using this module should thoroughly familiarize themselves with the participants handouts.

8.5 Module Duration

The Module is estimated to take a minimum of 5 hours 30 minutes.

8.6 Module Summary

Module 8: Banana Crop Health			
Sessions	**Training methods**	**Training materials**	**Time**
8.6.1 Introduction, objectives and expectations	• Personal introductions • Group work • Presentations • Plenary discussions	• Flips charts • Projector • Laptop	15 minutes
8.6.2 Major banana pests that cause economic losses and their management strategies	• Group work • Presentations • Plenary discussions • Practical exercise (arthropod pest specimens)	• Flips charts • Projector • Participants handouts • Laptop	1hour 30 minutes
8.6.3 Major banana diseases and nematodes that cause economic losses, conditions that favor their development and integrated management strategies	• Group work • Presentations • Plenary discussions • Practical exercise-Diseased plant specimens	• Flips charts • Projector Participants Handouts • Laptop	1 hours 30 minutes
8.6.4 Major weeds of economic importance in banana production and their management	• Group work • Presentations • Plenary discussions • Practical exercise-weed specimens	• Flips charts • Projector Participants handouts • Laptop	1 hour 30 minutes

8.6.5 Safe use of pesticides (insecticides, fungicides, herbicides) and updated source for registered pesticides	• Presentations • Practical demonstration • Plenary discussions	• Projector Flip charts • Participants handouts • Laptop	1 hour 30 minutes
8.6.6 Module Review	• Discussions/ Recap of module • Take away messages	• Flips charts • Sharing of presentations	15 minutes
Total			**6 hours 30 minutes**

8.7 Facilitator's Guidelines

Module 8: Banana crop health	
8.7.1. Introduction and levelling of expectations and objectives (30 minutes)	**Session Guide**
Introduction (15 minutes) *(The facilitator introduces the module and invites trainees to introduce themselves and state their expectations. The facilitator presents modules learning outcomes and expectations)* **Module Objectives (15 minutes)** *(The facilitator presents modules objectives)* By the end of the module trainee should be able to: • Identify major banana pests. • Appreciate sustainable integrated banana pest management (IPM) practices and scouting for threshold determination. • Appreciate pre-disposing factors of major banana diseases and nematodes. • Identify symptoms for specific diseases and nematodes common in banana producing areas. • Acquire skills in Integrated Disease Management (IDM) of banana. • Identify major weeds of economic importance in banana and their management. • Acquire knowledge on safe use of pesticides and source of up to date registered pesticides.	• Summarize traines' "expectations" and display. • PowerPoint Presentation • Distribute Participants Handouts on Module Objectives and Training Program

<table>
<tr><th>8.7.2. Major banana pests and their management strategies (1 hour 30 minutes)</th><th>Session Guide</th></tr>
<tr><td>

(The facilitator presents banana pests of economic importance and their management)

Group work (30 minutes)

- Trainees avail information on banana pests and management information from their counties

Plenary Presentation (15 minutes)

- Pest names and descriptions
- Emerging/migratory pests
- Signs of their infestation/type of damage
- Data on losses caused by the pests, scouting, threshold determination; (including, emerging/ migratory pests

Practical session (45 minutes)

- Identification of banana pests specimens.
- Demonstration of management strategies; Use of clean planting materials; Paring; hot water treatment; selection of mother plants for micro propagation; Sterilization of propagation/ transplanting media; hygiene

Discussion (10 minutes)

- Let the trainees to recall what they learned and discuss any issues that may arise.

</td><td>

- PowerPoint Presentation
- Group Work
- Practical Session
- Distribute Participants Handouts
- Brochures, leaflets and manual
- Printed photos of various pests, brochures

</td></tr>
<tr><th>8.7.3. Major banana diseases and nematodes, conditions that favor their development and their management strategies (1hour 30 minutes)</th><th>Session Guide</th></tr>
<tr><td>

(The facilitator presents banana diseases and nematodes of economic importance and their management)

Group work (20 minutes)
Determine banana diseases in specific counties

Plenary Presentation (30 Minutes)
Presentations on banana diseases and nematodes and conditions that favour their development

Practical / Exercise (30 Minutes)

- Identification of major banana diseases and nematode damage from samples presented.
- Demonstration of management strategies; clean planting materials; Paring; hot water treatment; selection of mother plants for micro propagation; Sterilization of propagation / transplanting media; field hygiene.

Discussion (10 minutes)

- Let the trainees recall what they learned and discuss any issues that may arise.

</td><td>

- PowerPoint Presentation by representative group leaders
- Distribute Participants Handouts (brochures, fact sheets, leaflets or printed photos of the major diseases and nematode damage
- Disease and nematode identification guidelines
- Live samples/printed photos of banana roots affected by nematodes

</td></tr>
</table>

8.7.4 Major weeds of economic importance in banana production and their management (1 hour 30 minutes)	Session Guide
(The facilitator presents weeds of economic importance and their management) **Group work (15 minutes)** • Determine weeds of economic importance in banana production systems in specific counties **Plenary Presentation (20 minutes)** • Weeds of economic importance in banana • Production systems in Kenya and their management. • Emerging weeds • Integrated weed management (IWD).Yield losses attributed to weeds. • Environmental/human factors affecting weeds **Practical Exercise (45 minutes)** • Identification of weeds from live specimens • Identification of weeds that cause economic losses in banana production. • Demonstration of management strategies **Discussion (10 minutes)** • Let the trainees recall what they learned and discuss any issues that may arise.	• PowerPoint Presentation by representative group leaders • Distribute Participants Handouts (brochures, leaflets or printed photos of the major weeds • Weed identification guidelines • Printed photos/ live samples of the major weeds

8.7.5. Safe use of pesticides and updated source for registered pesticides (1hour 30 minutes)	**Session Guide**
(The facilitator presents importance of safe use of pesticides) **Plenary presentation (30minutes)** • Presentation on safe use of pesticides (insecticides, fungicides, nematicides, herbicides). • Field scouting for arthropod pests, diseases and nematodes. • Threshold determination for arthropod pests, diseases and nematodes. **Practical (45minutes)** • Practical on updated PCPB registered list of products. • Mixing of pesticides, PHI, ITK products; and safety guidelines and precautions. **Discussions (15 minutes)** • Let trainees ask questions on any of the covered topical issues and critical areas to share with farmers on safe use of pesticides.	• PowerPoint Presentation by facilitator and representative group leaders • Demonstration of proper use of knap sack sprayer, protective gear and calibration of pesticides, sourcing for registered pesticide information online: on PCPB website • Distribute Participants Handouts (brochures, leaflets and manuals) • Disease management guidelines
8.7.6. Module review (15 minutes)	**Session Guide**
(The facilitator leads the trainees in reviewing the module) Summarize the main points of the training Together with the trainees review the main points on pests, diseases and weeds affecting banana production, their management and safe use of pesticides **(20 minutes)** *(Discuss with trainees the new things learnt from this Module. What are some of the issues that need clarification).*	• The last Participants' Handouts • Summarize the main points from the module on a flip chart and display

8.8 Participants' Handouts and reference materials

1. Pest identification and control factsheets
2. Disease identification and control factsheet
3. Weed identification and control factsheet
4. Fact sheet on propagation

References

1. Pest Control products registered for use in Kenya (2019). ttp://cacafrica.com/Public/Home/xiazai/PCPB%20List%20of%20Registered%20Products%20Version%201_2019.pdf
2. Banana Weevil. Infonet Biovision. https://www.infonet-biovision.org/PlantHealth/Pests/Banana-weevil Procured training materials for farmer groups
3. Banana Agribusiness: How it works. You tube video https://www.youtube.com/watch?v=oRr6n1pw9Ik

4. Banana Weevil. Infonet Biovision. https://www.infonetbiovision.org/PlantHealth/Pests/Banana-weevil

MODULE 9: POSTHARVEST TECHNOLOGIES

9.1 Introduction to the Module

Post-harvest food losses (PHL) refer to "*food losses –in quality and quantity– that occur between harvest and human consumption*". (i) Quantitative PHL refers to physical losses due due to spillage, mechanical damage, poor handling and limited storage capacity and (ii) Qualitative PHL refers to loss in desirable attributes such as loss of value due to the product being rendered unfit for its intended purpose as a result of decay or contamination. Banana is a highly perishable commodity. In Kenya, annual postharvest losses in fresh horticultural crops like bananas are estimated at more than 50% due to poor handling and lack of utilization of improved postharvest technologies. The postharvest losses in banana value chain is estimated to cost the country about KES 5.6 billion annually.

Reducing PHL will save money for farmers, businesses, and households that depend on the banana value chain and alleviate pressure on water and land, and other resources used in banana production. This will improve individual, household and business incomes along the banana value chain. This module introduces service providers, lead farmers and facilitators postharvest losses and management practices for minimizing the losses and enhancing quality and safety of banana for target groups.

9.2 Module Learning Outcomes

By the end of the module, the following should be achieved:

- The goal of postharvest management of banana shared and appreciated.
- Various postharvest tools and equipment for maturity evaluation for the different banana varieties identified and utilized.
- Best harvesting practices for quality bananas understood and shared.
- Postharvest and the whole range of postharvest handling practices for banana appreciated and understood.
- Constraints in banana postharvest chain discussed and shared.
- Banana postharvest losses (quantity and quality), causes, and their link to greenhouse gas emissions appreciated and shared.
- Climate smart postharvest TIMPs and reduction of postharvest losses demonstrated.
- Banana postharvest strategy for the priority opportunities demonstrated.

9.3 Module Target Group

This module targets agricultural extension service providers based at sub-county and Ward level. It can also be useful for private extension service providers and lead farmers.

9.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The facilitator using this module should thoroughly familiarize themselves with the participant's handouts.

9.5 Module Duration

The Module is estimated to take a minimum of 5 hours 30 minutes divided into 40% theory and 60% practical.

9.6 Module Summary

Module 9: Postharvest Management			
Sessions	**Training Methods**	**Training Materials**	**Duration**
9.6.1 Introduction, objectives and expectations	• Personal introductions • Group work • Presentations • Plenary discussions	• Projector • Flip charts • Laptop	15 minutes
9.6.2. Importance of postharvest management	• Presentations • Group work • Plenary discussions • Presentations • On-farm demonstrations	• Flips charts • Felt pens • Projector, Videos • Handouts • Laptop	30 minutes
9.6.3. Banana postharvest operations	• Group work • Brainstorming sessions • Presentations • Plenary discussions • On-farm practical demonstrations	• Flip charts • ProjectorVideos • Participants • Materials for demos (banana, baggings, etc.; refractometers, tools for maturity indices) • Handouts	45
9.6.4. Constraints in postharvest handling of banana, and suggested solutions	• Group discussions • Presentations • Plenary discussions	• Flip charts • Projector, Videos • Laptop	30 minutes

Module 9: Postharvest Management			
9.6.5. Banana post-harvest losses and the economic importance (food security, and link to greenhouse gas emissions)	• Presentations • Plenary discussions	• PowerPoint presentation • Participants Handouts • Videos	15 min
9.6.6 Climate smart Postharvest TIMPs to reduce losses, and attain best quality, higher prices: • Harvesting (maturity indices, bunch bagging, harvesting tools and assembly) • Sorting and grading • Curing and washing • Packaging • Low cost storage systems (Zero Energy Brick Cooler/ Coolbot™)	• Group work • Brainstorming sessions • Presentations • Plenary discussions • On-farm practical demonstrations	• PowerPoint presentation • Participants Handouts • Materials for demos (banana, baggings, panga, maturity indices tools and equipment, • Banana fruit protection bags, banana bunch covers (validation of the best polythene to use (colour and perforations) for bagging to give optimal results under different AEZs), etc.) • Video • Zero Energy Brick Cooler/ CoolbotTM), plastic crates • panga, gumboots	3 Hours
9.6.7. Module review	• Presentations • Plenary discussions	• Flips charts • PowerPoint • Module evaluation forms • Handouts	15 minutes
TOTAL			**5 hour 30 minutes**

9.7 Facilitator's Guidelines

Banana Postharvest Management	
9.7.1 Introduction, objectives and expectations (15 min)	**Session Guide**
Introduction and expectations (5 minutes) *(The facilitator introduces the module and invites participants to introduce themselves and state their expectations. The facilitator presents modules learning outcomes and expectations).* **Module Objectives (10 minutes)** *(The facilitator presents module objectives)* By the end of the module the trainee should be able to: • Share and appreciate goal of postharvest management of banana. • Identify and utilize various postharvest tools and equipment for maturity evaluation for the different banana varieties. • Share and carry out best harvesting practices for quality bananas. • Appreciate the meaning of postharvest and the whole range of postharvest handling practices for banana. • Discuss and share constraints in banana postharvest chain, and suggested solutions. • Appreciate and share banana postharvest losses (quantity and quality), causes, and their link to greenhouse gas emissions. • Conduct demonstrations (on-site and hands-on exercises) to farmers, traders and marketers on climate smart postharvest TIMPs for bananas on how to reduce postharvest losses, maintain quality and market value, increase shelf life and incomes.	• Summarize trainers "expectations" and display • PowerPoint presentation • Distribute participants' handouts on module objective and training program
9.7.2 Importance of postharvest management (30 minutes)	
Group work (20 minutes) Trainees discuss in groups, the importance of postharvest management in banana value chain. Trainees present in plenary, the results of group discussions. ***PowerPoint presentation (10 minutes)*** The facilitator presents on the importance of postharvest management in the banana value chain.	• PowerPoint presentation • Distribute handouts to participants • Brochures, leaflets and manual

9.7.3 Banana postharvest chain (1 hour)	
Group work - Discussion and plenary (30 minutes) Trainees discuss activities in banana postharvest chain in the respective counties / regions. Trainees present results of group work in the plenary. **PowerPoint presentation (15 min)** PowerPoint presentation on the operations in the banana postharvest chain, highlighting on: **Video presentation (15 min)** Video presentation on the banana postharvest value chain.	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual
9.7.4 Constraints in postharvest handling of bananas, and suggested solutions (30 minutes)	

• ***Group work - discussion and plenary (30 min)***
    o Trainees discuss constraints in the postharvest handling of banana, and suggested solutions
    o Trainees discuss results of group work in plenary
	• PowerPoint Presentation
	• Distribute Participants Handouts

9.7.5 Banana Postharvest Losses and Their Economic Importance (15 minutes)	**Session guide**
PowerPoint presentation (7 minutes) (*The facilitator should describe the postharvest losses and its impact on the value chain – food security and link to greenhouse gas emission*) Reduction in postharvest losses lead to • Increased market share and competitiveness of smallholders. • Stimulate growth of agribusiness industries, such as technology suppliers and logistics providers. • Generate more employment and income opportunities and stimulate the rural economy. • Promote gender equality as more women are involved in postharvest and marketing operations. Banana postharvest losses (quantity, quality) and their economic implications in banana production, consumption and marketing; link of postharvest losses and greenhouse gas emissions. **Discussion (8 minutes)** After the presentation demonstration, the facilitator will allow trainees to recall what they learned and discuss emerging issues	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual

<table>
<tr><th>9.7.6 Theory and Practical aspects of Postharvest operations of Banana (3 hours)</th><th>Session guide</th></tr>
<tr><td>Group work (30 min)
Trainees discuss in groups the following:
• Maturity indices for the different banana varieties
• Harvesting methods

Plenary presentation (20 min)
Trainees present findings of their group activity in the plenary

PowerPoint presentation (15 minutes)
• Maturity indices
• Bunch bagging
• Harvesting and assembly

Video presentation (15 minutes)
Video presentation on the banana postharvest value chain

On-farm practical demonstrations (1 hour 25 minutes)
Take the trainees through in field or on-farm demonstration on:
• Harvesting
• Sorting and grading
• Curing and washing
• Packing in crates
• Ripening chambers for immediate market
• Low cost storage systems (Zero Energy Brick Cooler / Coolbot™)

Discussion (15 minutes)
Guide the trainees to recall what they learned and discuss issues that may arise.</td><td>Distribute handouts to participants
• Manuals on postharvest operations of banana
• Brochures
• Posters
• Leaflets
• Demonstration material

Demonstration on some of the practices</td></tr>
</table>

9.7.7 Postharvest Management Strategy (30 minutes)	Session guide
PowerPoint presentation (15 minutes) *Facilitator presents the following on power point slides and flip charts:* • Identifying the product(s) of interest and its qualities • Consumer testing • Financial Analysis • Location, premises, Equipment (s) and production plan • Packaging and labelling • Regulations and regulatory agencies • Marketing plan **Discussion (15 minutes)** After the presentation, guide the trainees to recall what they learned and discuss issues that may arise.	Distribute handouts to participants Have a handout of a case study of a specific product. Discuss the task in groups and come with a presentation for plenary discussions/ inputs Q & A sessions
9.7.7 Model review (15 minutes)	**Session guide**
The facilitator leads the trainee in Summarizing the main points of the training. Together with the trainees review the main points about postharvest management of bananas. • What new things did you learn from this module? • What are some of the problems and issues that you have become more aware of in the module? • Have the trainees expectations been met	Group discussions Q & A session Module evaluation forms • Recap the main points • Test understanding • Link to the way forward • Participatory evaluation of the session

9.8 Participant's Handouts (Training and Reference Materials)

1. Training of Postharvest Trainers and Extension Specialists: Small-scale Postharvest Handling Practices and Improved Technologies for Reducing Food Losses. Lisa Kitinoja, the Postharvest Education Foundation. November 2016
2. Hand-outs and leaflets

MODULE 10

VALUE ADDITION IN BANANAS

This module outlines the learning outcomes, the category of trainees targeted, module duration, users and summary, facilitator's guidelines and participants' handouts.

10.1 Introduction to the Module

Malnutrition is a major problem in Kenya, with stunting levels estimated at 10% in children under 5 years Kenya has a great germplasm diversity of over 200 banana varieties, but only 10 are commercially exploited. Banana is highly nutritious and is a rich source of micro- and macro-nutrients. It can be used to address the malnutrition challenges. However, lack of information on the nutritional profile of the great germplasm diversity in banana, versatility of banana as a food ingredient and its potential in providing macro- and micronutrients (provitamin A carotenoids), has resulted in under-exploitation of the diverse varieties. Inappropriate cooking and preparation methods result in nutrient losses. Kenya has adopted a policy on compositing the staple food (maize) with other foods to reduce the pressure on maize occasioned by climate change–related phenomena such as pests and diseases. Banana is a promising compositing candidate crop to enhance human nutrition, through value addition. This module introduces service providers, lead farmers and trainees to the importance of banana in addressing food and nutrition security at household level, community level and industrial level. The module also covers constraints in value addition and consumption of banana and their suggested solutions, and various banana value added products that can be made at the household level which, if adopted, can help increase the banana consumption and utilization at the household level. TIMPS for community level and industrial processing of banana are also discussed.

10.2 Module Learning Outcomes

By the end of the module, the following should be achieved:

- Role of banana as a food and nutrition security crop appreciated and understood.
- Nutritional composition of banana, and impact of consumption on health, food security and income discussed and understood.
- Constraints in value addition and consumption of banana identified and suggested solutions made.
- Making several value added banana products (flour, wine, jam, juice, etc.) demonstrated.
- Value addition strategy for the priority opportunities discussed and shared.

10.3 Module Target Group

This module targets agricultural extension service providers based at sub county and ward level. It can also be useful for private extension service providers and lead farmers.

10.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The facilitator using this module should thoroughly familiarize themselves with the participant's handouts.

10.5 Module Duration

The Module is estimated to take a minimum of **5 hours,** divided into 40% theory and 60% practical.

10.6 Module Summary

Sessions	Training methods	Training Materials	Time
10.6.1. Introduction, objectives and expectations	• Personal introductions • Group work • Presentations • Plenary discussions	• Projector • Flip charts • Laptop	15 minutes
10.6.2. Role of banana as a food and nutrition security crop	• Presentations • Group work • Plenary discussions	• Projector, Participants Handouts	30 minutes
10.6.3 Banana nutritional composition and its role in human health	• Presentations • Plenary • Discussions • Group work • Demonstrations	• Projector, Participants Handouts • laptop	30 minutes
10.6.4 Constraints in value addition and consumption of banana, and suggested solutions	• Group discussions • Presentations • Plenary discussions	• Flip charts • Projector • Laptop	30 minutes
10.6.5. Value added banana products • Banana Flour/Powder • Bakery products derived from composite banana flour • Banana Chips • Banana jam and Jelly • Banana juice • Banana juice/ Unfermented beverages • Banana fermented beverages	• Presentations • Group • Discussions • Hands-on practical demonstrations • Field visit to processing firms / groups	• PowerPoint presentation • Participants Handouts / recipes • Assorted value addition equipment and ingredients • Sensory evaluation forms • Projector • Laptop	2 hours 30 minutes

10.6.6 Developing value addition strategy • Identifying the product(s) of interest and its qualities/ formulation • Consumer testing • Financial Analysis • Location, premises, Equipment (s) and production plan • Packaging and labelling • Regulations and regulatory agencies • Marketing plan	• Group discussions • Presentations • Plenary discussions	• Projector, Participants Handouts / recipes • Assorted value addition equipment and ingredients • Sensory evaluation forms	30 minutes
10.6.7. Module review and discussion	• Discussions	• Projector • Presentations • Have the expectation been achieved? • Module evaluation forms • Laptop	15 minutes
TOTAL			**5 hours**

10.7 Facilitator's Guidelines

Module 10: Value Addition of Banana	
10.7.1 Introduction, objectives and expectations (15 min)	**Session Guide**
Introduction and expectations (7 min) *(The facilitator welcomes participants to the module, then invites them to introduce themselves and state their expectations –).* **Module Objectives (8 min)** **(***The facilitator presents the objectives of the module***)** By the end of the module trainee should be able to: ▪ Recognize the role of banana as a food and nutrition security crop. ▪ Appreciate the nutritional composition of banana, and impact of consumption on health, food security and income. ▪ Identify the constraints in value addition and consumption of banana, and suggested solutions. ▪ Demonstrate making value added banana products (flour, wine, jam, juice, etc.) ▪ Discuss and share how to develop a value addition strategy for the priority opportunities.	Summarize trainers "expectations" and display Distribute handouts to participants • Module objectives • Program

10.7.2 Role of banana as a food security and nutritional crop (45 minutes)	Session Guide
(The facilitator presents malnutrition cases in Kenya and importance of banana in addressing food security and malnutrition challenges) **Plenary Presentation (15 minutes)** • Micronutrient malnutrition cases in Kenya. • Dietary nutrient requirements (focusing on VMGs), food nutrient deficiency symptoms, sources of nutrients and their implication in human health. **Group work and discussion (30 min)** Trainees discuss in groups, the main malnutrition challenges in their respective counties / regions. Let the trainees recall what they learned and discuss any issue that may arise.	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual, factsheets, posters
10.7.3. Banana nutritional composition and their intake impact in human health (30 minutes)	**Session Guide**
Group work (10 min) Trainees discuss in groups: • The constraints in consumption of banana, and suggested solutions. • Health benefits of banana consumption. **Plenary Presentation and Discussion (10 min)** Trainees present results of group work in the plenary **PowerPoint Presentation (10 minutes)** • Banana nutritional composition and health benefits. • Nutrient guidelines and nutritional requirements for different VMGs. Let the trainees recall what they learned and discuss any issues that may arise.	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual, factsheets, posters

10.7.4. Banana-based value added products (2 hours 30 minutes)	Session Guide
Group work, discussion and plenary (30 min) • Trainees discuss constraints in value addition of banana, and suggested solutions (at household level, farmers group level, and industrial level). **PowerPoint Presentation (10 min)** • Meaning of value addition. • Importance of value addition. • Requirements for value addition of banana. • Banana-based value added products; and • sensory evaluation of value added banana products. **Group work (30 min)** In groups, trainees discuss the following: • Household level utilization of bananas. • Community / farmers group level utilization of banana. • Industrial level utilization of banana. • Value added banana products already in the market. • Possible / potential banana-based value added products. **Field visit, optional (1 hour)** {*optional, depending on availability of the firms/groups*} trainees visit banana processing firms / groups **Practical exercise and Discussion (30 minutes)** • Hands-on practical demonstration on formulation of banana value added products. • Practical on sensory evaluation of banana value added products. After the practical demonstration, the facilitator will allow trainees to recall what they learned and discuss emerging issues.	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual, • Recipes • Sensory evaluation forms • Assorted value addition equipment and ingredients Hands-on Practical Exercise (making value added products, sensory evaluation of the products) Field visit to processing firms and groups Practical demonstration

10.7.5 Value Addition Strategy (30 minutes)	Session guide
PowerPoint presentation (15 minutes) *Facilitator presents the following on power point slides and flip charts:* • Identifying the product(s) of interest and its qualities/ formulation • Consumer testing • Financial Analysis • Location, premises, Equipment (s) and production plan • Packaging and labelling • Regulations and regulatory agencies • Marketing plan **Discussion (15 minutes)** After the presentation, guide the trainees to recall what they learned and discuss issues that may arise.	Distribute handouts to participants Have a handout of a case study of a specific product. Discuss the task in groups and come with a presentation for plenary discussions/ inputs Q & A sessions
10.7.6 Model review (15 minutes)	**Session guide**
The facilitator leads the trainees in summarizing the main points of the training. Together with the trainees review the main points about value addition of bananas: • What new things did you learn from this module? • What are some of the problems and issues that you have become more aware of in the module? • Take home messages.	Group discussions Q & A session ▪ Recap the main points ▪ Test understanding ▪ Link to the way forward ▪ Participatory evaluation of the session ▪ Module evaluation forms

10.8 Participant's Handouts (Training and Reference Materials)

1. Mohapatra, D., Mishra, S., Singh, C.B. and Jayas, D.S., 2011. Post-harvest processing of banana: opportunities and challenges. *Food and Bioprocess Technology*, *4*(3), pp.327-339.
2. Anyasi, T.A., Jideani, A.I. and Mchau, G.R., 2013. Functional properties and postharvest utilization of commercial and noncommercial banana cultivars. *Comprehensive Reviews in Food Science and Food Safety*, *12*(5), pp.509-522.

MODULE 11
MECHANIZATION OF BANANA PRODUCTION

11.1 Introduction to the module

Agricultural mechanization supports enhancement of production, productivity and profitability in agriculture by achieving timeliness in farm operations. It comes along with precision in metering and placement of inputs, reducing available input losses, increasing utilization efficiency of costly inputs (seed, chemical, fertilizer, irrigation, water etc.), reducing unit cost of produce, enhancing profitability and competitiveness in the cost of operation. It also helps in the conservation of agricultural produce and byproducts from qualitative and quantitative damages; enables value addition and establishment of agro processing enterprises for additional income and employment generation from farm produce. Agricultural mechanization is one of the important inputs that has potential to revolutionize all round development in the rural Kenya.

11.2 Module learning outcomes

By the end of the module the following should be achieved:

- Climate smart tillage operations appreciated.
- Calibration of fertilizer and seed rate for planters applied.
- Weed control equipment and tools, usage demonstrate.
- Harvest timing, yield estimation, machines and tools appreciated.
- Estimation of pre-harvest and harvesting losses demonstrated.
- Machine usage and procedure for banana grading demonstrated.

11.3 Module Target Group

This module is intended for private service providers and county public extension agents and lead farmers.

11.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The facilitator using this module should thoroughly familiarize themselves with the participant's handouts.

11.5 Module Duration

The Module is estimated to take a minimum of 2 hours

11.6 Module Summary

Module 11. Mechanization of Banana production activities			
Sessions	**Training methods**	**Training materials**	**Duration**
11.6 .1 Introduction, objectives and expectations	• Personal introductions/know your audience • Presentations • Plenary discussions	• Flips charts • Projector • Laptop	15 minutes
11.6.2. Banana Climate smart tillage and hole preparation	• Presentations • Plenary discussions	• Flip chart • Projector • Participants Handouts • Laptop	30 minutes
11.6.3. Use of pest control implements and tools	• Presentations • Plenary discussions	• Flip chart • Power Point presentation • Participants Handouts • Laptop • Projector	30 minutes
11.6.4. Techniques and methods of using weed control equipment in Banana	• Presentations • Plenary discussions	• Flip chart • Projector • Participants Handouts • Practical • Laptop	30 minutes
11.6.5. Module review	• Presentations • Plenary discussions	• Flip chart • Projector, Participants Handouts • Practical • Laptop	15 minutes
Total			2 hours

11.7 Facilitator's Guidelines

<table>
<tr><th colspan="2">Module 11.7.1: Mechanization of Banana production activities</th></tr>
<tr><td>11.7.1 Introduction, Objectives and Expectations (15 minutes)</td><td>Session Guide</td></tr>
<tr><td>(The facilitator welcomes s the trainees to the module. They are then invited to introduce themselves and state their expectations).

Module Objectives (15 minutes)

(The facilitator presents modules objectives)

By the end of the module the trainee should be able to:
• Appreciate various climate smart tillage operations.
• Apply various calibration of fertilizer and seed rate for planters.
• Demonstrate weed control equipment and tools, usage.
• Appreciate harvest timing, yield estimation, machines and tools.
• Show estimation of pre-harvest and harvesting losses.
• Demonstrate machine usage and procedure for banana grading.</td><td>Summarize trainees' "expectations" and display on a flip chart.

• PowerPoint Presentation
• Distribute Participants Handouts on Module Objectives and Training Program</td></tr>
<tr><td>11.7.2 Banana climate smart land preparation tools (30 minutes)</td><td>Session Guide</td></tr>
<tr><td>(The facilitator presents on the banana merchandisable components of economic importance)

Plenary Presentation (15 Minutes)
• Overview of the banana mechanization activities
• Climate smart tillage options

Discussion (15 Minutes)

Let the trainees recall what they learned and discuss any issues that may arise.</td><td>• PowerPoint Presentation
• Distribute Participants Handouts
• Brochures, leaflets and manual
• All participants</td></tr>
</table>

11.7.3 Use of pest control implements and tools (30 minutes)	Session Guide
Plenary Presentation (15 minutes) • Calibration of fertilizer/manure in planting **Discussion (15 Minutes)** Let the trainees recall what they learned and discuss any issue that may arise.	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual
11.7.4 Use of weed control implements and tools (30 minutes)	**Session Guide**
Plenary Presentation (15 minutes) • Techniques and methods of using Banana pest/disease/weed control equipment's (Sprayers) **Discussion (15 Minutes)** Let the trainees recall what they learned and discuss any issue that may arise.	• PowerPoint Presentation • Distribute Participants Handouts • Brochures, leaflets and manual
11.7.5 Module review (15 minutes)	**Session Guide**
The facilitator leads the trainees in reviewing the module) Summarize the main points of the training and together with the trainees review the main points: • Climate smart tillage options • Calibration of fertilizer in planting • Use of pest control implements and tools • Machine and procedure for grading • Discuss with trainees the new things learnt from this Module. What are some of the problems and issues that they have become more aware of in the module?	• The last Participants' Handouts • Summary of the main points on from the module on a flip chart and display

11.8 Participant's Handouts (Training and Reference Materials)

Provide necessary materials

MODULE 12

BANANA BUSINESS AND MARKETING

12.1 Introduction to the Module

This module introduces service providers, lead farmers and trainees to the importance of banana farming business and marketing. The module also focuses on helping farmers to acquire sustainable business skills that will assist them to build a long-term financial resilience to the negative impact of increasing drought. This would be achieved through production and development of marketable banana products that contribute to addressing improved food and nutritional security and better income at the household level. The areas covered under this module include the concept of banana farming business, opportunities of banana products in local and international markets, risk management in banana farming business, market analysis and importance of a SWOT analysis, farm decision analysis tools (partial budget, break-even and gross-margin) and business plan and marketing of banana farm produce and products.

12.2 Module Learning Outcomes

By the end of this module training, the following should be achieved:

- The importance and benefits of subsistence and commercial banana farmingappreciated and understood.
- The risks and opportunities in banana production identified and appreciated.
- The general principles for record keeping and elements and the principles of preparing and implementing a profitable **Business plan** mapped and shared.
- Types and uses of budgets and profitability analysis identified and shared.
- Vital elements for conducting **SWOT** analysis matrix in marketing banana produce and products shared and appreciated.
- The principles of 5 ‘Ps’ in marketing banana and delivery systems discussed and understood.
- E-marketing and contract Farming discussed.

12.3 Module Target Group

This module is intended for private service providers, public extension agents and lead farmers in the banana producing counties.

12.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The trainer using this module should thoroughly familiarize themselves with the required participant's handouts.

12.5 Module Duration

The module is estimated to take a minimum of 4 hours.

12.6 Module Summary

Module 12: Banana Business and Marketing			
Sessions	**Training methods**	**Training materials**	**Time**
12.6.1 Introduction, learning expectations and outcomes	• Personal introductions • Group discussions • Plenary discussions • Presentations	• Flips charts • Felt pens • Projector	15 minutes
12.6.2 Subsistence and Commercial Banana Production	• Presentations • Plenary discussions	• Flips charts • Felt pens • Projector • Participants Handouts	20 minutes
12.6.3 Risk management in banana farming business, Opportunities and Participation requirements for Banana produce and products in local and international markets	• Presentations • Plenary discussions • Demonstrations	• Flips charts • Felt pens • Projector • Participants Handouts	30 minutes
12.6.4 Banana Farm Business Record Keeping and developing a Business plan for planning	• Presentations • Plenary discussions	• Flips charts • Felt pens • Projector • Participants Handouts	1 hour
12.6.5 Managing banana business performance by developing Partial budget, Break-even, and Gross margin analysis	• Presentations • Practical exercises • Plenary discussions	• Flips charts • Felt pens • Projector • Participants Handouts	40 minutes

12.6.6 Marketing of banana produce and products by developing Strengths, Weaknesses, Opportunities and Threats (SWOT) matrix, use of principles of 5 'Ps', Contract Farming and E-marketing	• Presentations • Practical exercise • Plenary discussions	• Flips charts • Felt pens • Projector • Participants Handouts	1 hour
12.6.7. Module Review	• Plenary discussion	• Flip charts • Felt pens	15 minutes
Total			**4 hours**

12.7 Facilitator's Guidelines

Module 9: Banana Business and Marketing	
12.7.1 Introduction and Expectations (15)	**Session Guide**
(The facilitator welcomes trainees to the module. They are then invited to introduce themselves and state their expectations) **Participants Expectations (7 minutes)** **Module Objectives (8 minutes)** *(The facilitator presents module learning objectives)* **By the end of this module, the trainee should be able to:** • Appreciate the banana business concept. • Distinguish between subsistence and commercial farming practices. • Identify risks in banana farming business and opportunities for banana products and requirements for participating in local and international markets. • Outline the importance of record keeping and common types of records required in a banana farm business. • Appreciate the principles of preparing and implementing a profitable **Business plan** • Explain the concept of farm budgeting, different types of farm budgets and the components of a farm budget. • Define the concept of marketing and approaches (SWOT, 5Ps, E-marketing and Contract Farming. • Evaluate different target markets and marketing channels for banana produce and products.s	• Summarize trainees' "Expectations" and display. • PowerPoint Presentation • Distribute Participants Handouts on Module Objectives and Training Program

12.7.2 Subsistence and Commercial Banana Production (20 minutes)	Session guide
Plenary Presentation (10 minutes) • Concept of a banana business. • Key principles of banana business. • Commercialization and competitiveness of banana as a commodity crop (from subsistence to commercial farming). • Advantages of practicing banana farming as a profitable agro-enterprise. **Plenary Discussion (10 minutes)** Discussion the concept of banana farming as a business.	• PowerPoint presentation • Distribute participants handouts • Group Exercise
12.7.3 Risk management in banana farming business, Opportunities and Participation requirements for Banana produce products in local and international markets (30 minutes)	**Session Guide**
Plenary Presentation (10 minutes) • The concept of risk. • Risks associated with banana farming business • Risk management measure at farm level • Banana business environment both locally and internationally • Banana production, demand and supply trends • Diversity in marketable banana-based products • Banana products market quality requirements and their respective prices. **Plenary Discussion (20 minutes)** • Let the trainees recall what they learned and discuss any issue that may arise. • Discussion on banana products available locally and unexploited market opportunities.	• PowerPoint presentation • Distribute participants handouts

12.7.4 Banana Farm Business Record Keeping and developing a Business plan for planning (1 hour)	Session Guide
Plenary Presentation (15 minutes) • Why should records be kept in a banana farm business • General principles for record keeping in banana business • Why plan banana business? • Elements of a successful business plan • Guidelines and steps in preparing a competitive and profitable business plan • Characteristics of a good business plan. **Practical exercises (45 minutes)** *Trainees to develop a business plan for a banana business (1 hour)*	• PowerPoint presentation • Distribute participants handouts • Business Training Notes • Business planning Basics Factsheet • Practical Exercises
12.7.5 Managing banana business performance by developing Partial budget, Break-even, and Gross margin analysis (40 minutes)	**Session Guide**
Plenary Presentation (10 minutes) • What is banana farm budgeting? • Uses of banana farm budgets • Common types of farm budgets • Steps in preparing farm budgets: Partial budget analysis (in farm changes or alternatives), break-even analysis and gross margin analysis **Practical exercises (30 minutes)** *Trainees to develop a gross margin budget for a one acre banana business*	• PowerPoint presentation • Distribute participants handouts • Business Training Notes • Farm budgeting Factsheet • Practical Exercises

<table>
<tr><th>12.7.6 Marketing of banana products by developing Strengths, Weaknesses, Opportunities and Threats (SWOT) matrix, use of principles of 5 ‘Ps’, Contract Farming and E-marketing (1 hour)</th><th>Session Guide</th></tr>
<tr><td>Plenary Presentation (15 minutes)

• Understanding the marketing concept.
• How to conduct market survey for banana produce and products.
• Identification of target buyers and marketing channels.
• What S, W, O, and T mean for a banana business
• Describe a banana farming business by its strengths, weaknesses, opportunities and threats.
• Marketing options for banana products (5Ps) – marketing mix.
• Contract farming in banana business.
• Role of marketing institutions.

Practical exercises (45 minutes)

Group exercise on SWOT analysis on banana marketing

(The group then compares the list from the facilitator and that developed during the group exercise).</td><td>• PowerPoint presentation

• Distribute participants handouts

• Business Training Notes

• SWOT Factsheet

Practical Exercise</td></tr>
<tr><th>12.7.7 Module review (15 minutes)</th><th>Session guide</th></tr>
<tr><td>The facilitator leads the trainees in reviewing the module).

Summarize the main points of the training and together with the participants review the main points:

• Transformation process of banana farming from subsistence to commercial.
• Options for risk management in banana business.
• Opportunities for banana produce and products in local and international markets.
• Participation requirements in local and international markets.
• Importance of record keeping and business plan
• Tools for financial management.
• Marketing options for banana produce and products.

(Participants list the main points learnt during the training. Discuss with participants new things learnt from this Module. What are some of the problems and issues that they have become more aware of in the module?).</td><td>• Group discussions

• Q& A session

• Recap the main points

• Test understanding

• Participatory evaluation of the session</td></tr>
</table>

12.8 Participant's Handouts (Training and Reference Materials)

References

1. Luyayi, F. et al. (2014). Farmers training entrepreneurship manual. Published by the World Agroforestry centre. United Nations Avenue, Gigiri. ISBN: 978-92-9059-356-0. Website: www.worldagroforestry.org

2. USAID/Zimbabwe (2012). Farming as a Family Business. Training Manual. A guide to higher profits through managing farming as a family business.

3. Banana Agribusiness: How it works. You tube video https://www.youtube.com/watch?v=oRr6n1pw9Ik

MODULE 13

CROSS-CUTTING ISSUES

SUB-MODULE MODULE 13.1: AGRICULTURAL INNOVATION PLATFORMS

13.1.1 Introduction to the Module

This module exposes the service providers, lead farmers and trainees to an innovation system based configuration of stakeholders called the Agricultural Innovation Platform (AIP). It is an organizational model for stimulating innovation and development that brings actors together in a way that pools together skills and knowledge used to address challenges and utilize opportunities. The actors include individuals, private and public sector organizations, policy makers and other value chain stakeholders brought together to seek a solution to a challenge hindering agricultural productivity within a value chain such as bananas. The Agricultural Innovation Platform facilitates actors to interact, innovate, learn and change with time as they seek a solution to the common challenge or compelling agenda. In an innovation platform, information exchange takes place in an environment where every actor's contribution is valued and various benefits accrue to all in a win-win situation.

13.1.2 Module learning Outcomes

By the end of the training module, to the following should be achieved:

- The attributes of an innovation platform are identified and described.
- Ways of stakeholders mobilization for initiation of an Agricultural Innovation Platform demonstrated.
- The establishment, management and monitoring of Agricultural Innovation Platforms demonstrated.
- Process of building innovation capacity of the actors is described.

13.1.3 Module Target Group

The target users are public county extension officers, private agricultural service providers, and lead farmers.

13.1.4 Module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The trainer using this module should thoroughly familiarize themselves with the participants handouts.

13.1.5 Module Duration

The module is estimated to take a minimum of 2 hours

13.1.6. Module summary

Module Summary Sessions	Training methods	Training materials	Time
13.1.6.1 Introduction, objectives and expectations	• Personal introductions • Presentations • Plenary discussions	• Flips charts • Projector • Laptop	15 minutes
13.1.6.2 An overview of attributes of an Agricultural Innovation Platform	• Power point Presentations • Plenary discussions	• Flip charts • Projector • Participants Handouts • Laptop	15 minutes
13.1.6.3. Pre-formation stages –stakeholder mobilization and sensitization.	• Power point presentations • Plenary discussions • Role plays	• Flips charts • Projector • Handouts • Roles • Laptop	15 minutes
13.1.6.4 AIP Phases (Initiation, Establishment, Management and Sustenance)	• Power point presentations • Plenary discussions • Role plays • Video clip	• Flips charts • Projector • Handouts • Roles • Laptop	1 hours
13.1.6.5. Module review	• Discussions	• Flip Charts	15 minutes
Total			**2 hours**

13.1.7 Facilitator's Guidelines

<table>
<tr><th colspan="2">Module 13.1: Agricultural Innovation Platform</th></tr>
<tr><th>13.1.7.1 Introduction, levelling of expectations and objectives (20 minutes)</th><th>Session Guide</th></tr>
<tr><td>Introduction (10 minutes)

(The facilitator welcomes trainees to the module. They are then invited to introduce themselves and state their expectations).

Module Objectives (10 minutes)

(The trainee presents modules objectives and levels out expectations).

By the end of the module participants should be able to:
• Explain characteristics of an innovation platform.
• Mobilize and sensitize stakeholders.
• Describe how to initiate and establish Agricultural Innovation Platforms.
• Explain how to manage and sustain innovation capacity of actors in Agricultural Innovation Platforms.</td><td>• Summarize trainees' "Expectations" and display.
• PowerPoint Presentation
• Module Objectives and Training Program</td></tr>
<tr><th>13.1.7.2 The characteristics of an innovation platform (20 minutes)</th><th>Session Guide</th></tr>
<tr><td>(The facilitator should present an overview of innovation platforms and their main characteristics).

Plenary Presentation (10 minutes)

• Past progression of research and extension models and their shortcomings.
• Agricultural Innovation Systems perspective.
• and Agricultural Innovation Platforms model
• Comparison of Agricultural Innovation Platforms with social and technical events working through committees with different roles but common goals.
• Value chain actor linkages and other benefits.

Discussion (10 minutes)

Let the trainees recall what they learned and discuss any issues that may arise.</td><td>• PowerPoint Presentation
• Notes Handouts,
• Brochures, information leaflets and manuals</td></tr>
</table>

<table>
<tr><th>13.1.7.3 Preformation and formation phases of the Banana AIP (1 hour)</th><th>Session Guide</th></tr>
<tr><td>

Plenary Presentation (40 minutes)

Facilitator presents:

Initiation or preformation phase

- Engagement or mobilization of stakeholders in the banana value chain.
- Visioning process and rules of engagement mediated by an initiator such as an change agent.

Establishment

- Assessment of the status of the value chain to clearly identify the compelling agenda or bottleneck - APVC analysis to identify weaknesses in the chains.
- Laying out of proper plans to define roles, establish task- based committees, expected milestones and resourcing strategies.

Management

- Keeping stakeholders focused on the vision and upholding values to ensure an inclusive and transparent process.
- Neutral facilitation to ensure joint strategy building and action and the coordination of support activities.
- Managing emerging experts taking up leading roles and issues as champions.

Sustainability

- Guiding in evolving and identifying fresh issues or challenges
- Maintaining capacity acquired to address new issues or challenges in subsequent cycles.

Discussion (20 minutes)

Let the trainees recall what they learned and discuss any issues that may arise.

</td><td>

- PowerPoint Presentation
- Distribute Participants Handouts
- Brochures, Leaflets, Manuals
- Short video clips

</td></tr>
</table>

13.1.7.4. Module review (20 minutes)	Session Guide
(The facilitator leads the trainees in reviewing the module). Summarize the main points of the training and together with the trainees review the main points on: • AIP characteristics and initiation. • AIP establishment and management. • Sustenance of Banana AIPs. *(Discuss with trainees new things learnt from this Module. What are some of the problems and issues that they have become more aware of in the module?).*	• The last Participants' Handouts • Summarize the main points from the module on a flip chart and display

13.1.8 Participant's Handouts (Training and Reference Materials)

References:

1. Kamau, G.M. and Makini F.W. (2019). Agricultural Innovation Platforms for knowledge exchange and learning for technical, economic, social and institutional change

2. Felister Makini, Wellington Mulinge, Lawrence Mose, Beatrice Salasya, Geoffrey Kamau, Margaret Makelo, and On'gala, J. (2018). Impact of Agricultural Innovation Platforms on Smallholder livelihoods in Eastern and Western Kenya. FARA Research Results Vol2(6)

3. F. Makini, G. Kamau, M. Makello, A. Adekunle, G.Mburathi. (2013). Operational field guide for developing and managing local agricultural innovation platforms KARI ISSBN 978-9966-30-004-1

4. Gabriel Teno and Jean-Joseph Cadilhon (2016). Innovation platforms as a tool for improving agricultural production: the case of Yatenga province, northern Burkina Faso (Journal of field actions, vol 9, 2016)

SUB MODULE 13.2: GENDER MAINSTREAMING AND SOCIAL INCLUSION IN THE BANANA VALUE CHAIN

13.2.1 Introduction to the Sub-module

Banana is a major agro-enterprise which involves all the gender categories (men, women, youth vulnerable marginalized groups (VMGs) in various activities along its value chain from production, marketing and consumption. However, women perform most of the crop production activities such as weeding, harvesting and carrying the produce to the market while men mostly dig the planting holes. Despite this huge women's contribution, gender inequalities exist in all areas of the value chains. Some of these gender inequalities include: division of labour, access to and control of resources and decision making. These inequalities limit women, youth and VMGs access to and benefits from the various Technologies Innovations and Management Practices (TIMPs) at different nodes of the value chain. At the macro-level, effective participation of women and youth in groups and market activities is constrained by their low decision making power, lack of voice and lack of access to financial resources. Gender analysis examines the productive and reproductive roles of men and women; access, control and ownership of resources; levels of power relations; differential needs, constraints and opportunities; and impact of these differences (positive/ negative) on lives of men, women, youth and the VMGs.

Banana value chain TOT and TIMPs interventions, when designed and implemented with gender equitable principles, can foster adoption leading to increased productivity as well as enhanced social and environmental impacts.

The overall objective of this module is to empower the trainees with knowledge and skills to ensure that gender mainstreaming and social inclusion in banana TIMPs is enhanced by field agricultural practitioners and extension officers as an effort geared towards achieving Climate Smart Agriculture "triple wins" in target counties.

13.2.2 Sub-module learning outcomes

By the end of the training module, the following should be achieved:

- Enhanced understanding of the concept of gender mainstreaming and social inclusion in banana value chain.
- Increased appreciation of youth empowerment in banana value chain.
- Enhanced knowhow of women empowerment in banana value chain.
- Increased understanding of the strategies for inclusion of vulnerable and marginalized groups in banana value chain.
- Increased understanding of socio-cultural barriers in banana value chain.
- Environmental and social management framework (ESMF) tool appreciated and understood.

13.2.3 Sub-module Target Group

This module is intended for county public, private extension agents and lead farmers.

13.2.4 Sub-module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). This module outlines the learning outcomes, the category of trainees targeted, module summary, and participants' handouts. The facilitator using this module should thoroughly familiarize themselves with the participant's handouts.

13.2.5 Sub-module Duration

The Sub - module is estimated to take a minimum of 2 hours.

13.2.6 Sub-module Summary

Sub-module 13: Gender mainstreaming and social inclusions in the banana value chain			
Sessions	**Training methods**	**Training materials**	**Duration**
13.2.6.1 Introduction, expectations and objectives	• Personal introductions • PowerPoint Presentations • Plenary discussions	• Flips charts • Felt pens • Projector • Laptop • Participants handouts	15 minutes
13.2.6.2 Gender mainstreaming in banana value chain	• PowerPoint Presentations • Group discussions • Plenary discussions	• Flips charts • Felt pens • Projector • Participants handouts • Laptop	20 minutes
13.2.6.3 Youth empowerment in banana value chain	• PowerPoint Presentations • Group discussions • Plenary discussions	• Flips charts • Felt pens • Projector • Participants handouts • Laptop	15 minutes
13.2.6.4 Women empowerment in banana value chain	• PowerPoint Presentations • Group discussions • Plenary discussions	• Flips charts • Felt pens • Projector • Participants handouts • Laptop	15 minutes

13.2.6.5 Strategies for inclusion of vulnerable and marginalized groups in the value chain	• PowerPoint Presentations • Group discussions • Plenary discussions	• Flips charts • Felt pens • Projector • Participants handouts	20 minutes
13.2.6.6 Environmental and Social Management Framework **Environmental, social risks** **Environmental and social safeguards** **Socioeconomic Impacts**	• PowerPoint Presentations • Group discussions • Plenary discussions	• Flips charts • Felt pens • Projector • Participants handouts	20 minutes
13.2.6.7 Module Review	• Plenary discussions	• Flips charts • Felt pens	15 minutes
Total			**2 hours**

13.2.7 Facilitator's Guidelines

Module 10: Gender mainstreaming and social inclusion in banana value	
13.2.7.1 Introduction, Objectives and Expectations (15 minutes)	**Session Guide**
(The facilitator welcomes trainees to the module. They are then invited to introduce themselves and state their expectations). **Module Objectives (15 minutes)** (The facilitator presents the modules objectives) By the end of the module training the trainees is expected to:- • Appreciate gender main streaming and social inclusion in banana value chain. • Recognize need for youth empowerment in green banana value chain. • Understand women empowerment in banana value chain. • Identify strategies for inclusion of vulnerable and marginalized groups in banana value chain. • Understand socio-cultural barriers in banana value chain. • Identify the environmental and social management framework (ESMF) tool.	• Summarize trainees' "Expectations" and display. • PowerPoint Presentation • Flipcharts • Group exercise • Objectives and Training Program

13.2.7.2 Gender mainstreaming and social inclusion in banana value chain (20 minutes)	Session Guide
(The facilitator presents Gender mainstreaming in banana value chain) **Plenary Presentation (10 minutes)**: • Definition of gender • What is gender mainstreaming and why it is important. • Who does what? (gender division of roles in bananas value chain) • Who owns what? (access and control of resources & benefits). • Who makes which decisions? • Socio-cultural limitations related to banana value chain • Existing policies in gender mainstreaming. **Group exercise and General discussion** (10 minutes) Let the trainees recall what they learned and discuss any issues that may arise.	• PowerPoint Presentation, Group exercise • Plenary discussions • Distribute Participants Handouts • Group exercise • Plenary discussions
13.2.7.3 Youth empowerment in banana value chain (15 minutes)	**Session Guide**
(Facilitator presents on factors that lower participation of youth in agriculture and the leads trainees in groups discuss). **PowerPoint presentation (7 minutes)** • Why agriculture is not attractive to youth. • Youth's role in the value chain. • Strategies to empower youth in banana gram value chain. **Group work and Group discussion (8 minutes)** Let the trainees recall what they learned and discuss any issues that may arise.	• PowerPoint Presentation • Group exercise • Plenary discussion

13.2.7.4 Women empowerment in banana value chain (15 minutes)	Session Guide
(The facilitator presents on women roles, challenges and strategies of empowering women in their participation in the banana value chain. Give room for a discussion in groups to delve deeper into the issues) **PowerPoint Presentation (7 minutes)** PowerPoint presentation highlighting: • Women's role in the value chain. • Challenges facing women in the value chain. • Strategies for empowering women in the value chain. **Group exercise and Plenary discussion (8 minutes)**	• PowerPoint Presentation • Distribute participants handouts • Group exercise • Plenary discussion

13.2.7.5 Strategies for inclusion of vulnerable and marginalized groups in banana value chain (20 minutes)	Session Guide
(The facilitator guides through a presentation on strategies for inclusion of vulnerable and marginalized groups in banana value chain. There after the facilitator guides a group exercise discussing the strategies available). **Plenary Presentation (10 minutes)** PowerPoint presentation highlighting; • Who are vulnerable and marginalized groups (VMGs) • Why gender inequality exist • Social inclusion and why • Strategies of inclusion of VMGs **Group exercise** In groups trainees discuss the strategies available for VMG inclusion in Banana value chain **Plenary discussions** (10 minutes) Groups present summary of their discussions and facilitator guides in summarising findings	• PowerPoint Presentation • Group exercise • Plenary discussion

13.2.7.6 Environmental and social management framework (ESMF) (20 minutes)	Session Guide
(Facilitator guides on the principle of ESMF and identifies issues that need consideration. He/she then guides a discussion on the same) **Plenary Presentation (10 minutes)** PowerPoint presentation highlighting; • Objective of ESMF in banana value chain • Environmental, social risks • Environmental and social safeguards • Environmental and socioeconomic impacts of banana value chain activities **Group exercise and (10 minutes)** In groups trainees discuss issues that need consideration in their respective areas **Plenary discussion** Groups present summary of their discussions and facilitator guides in summarising findings	• PowerPoint Presentation • Group exercise • Plenary discussion
13.2.7.7 Module review (15 minutes)	**Session Guide**
(The facilitator leads the participants in reviewing the module) Facilitator summarizes the main points of the training and together with the trainees reviews the main points: • What is gender mainstreaming and why it is important. • Youth empowerment in banana value chain. • Women empowerment in banana value chain. • Strategies for inclusion of vulnerable and marginalized groups in banana value chain. • Socio-cultural barriers in the value chain. • Environmental and socioeconomic impacts of banana activities. Let the trainees recall what they learned and discuss any issues that may arise.	· Summary of the main points on from the module on a flip chart and display

13.2.8 Participant's Handouts (Training and Reference Materials)

SUB MODULE 13.3: CLIMATE-SMART AGRICULTURAL POLICY OPTIONS

13.3.1 Introduction to the sub-module

Kenya adopted Vision 2030 in 2007 as blue print and roadmap for political, social and economic development of the country in the next two decades. The Vision also identifies Agriculture as the engine of growth through transformation of smallholder and subsistence agriculture to innovatively and commercially oriented agriculture. Kenya promulgated the new constitution in 2010 which proposed two levels of governments (national and county) with clearly defined functions. Agriculture is one of the devolved governance functions. However, agriculture in Kenya is facing many challenges and threats such as climate change, declining agricultural performance, limited high potential agricultural land and over-reliance on rain fed agriculture, limited diversification of agricultural production, poor and inadequate rural infrastructure, inadequate and declining research in agriculture, agricultural sector financing and related activities and low technical capacity among the actors.

Therefore, agricultural policy in Kenya revolves around the main goals of increasing productivity and income growth, especially for smallholders; enhanced food security and equity, emphasis on irrigation to introduce stability in agricultural output, commercialization and intensification of production especially among small scale farmers; appropriate and participatory policy formulation and environmental sustainability. This sub module introduces the national and county governments, service providers, lead farmers, facilitators and relevant stakeholders in the design and implementation of effective climate-smart-sensitive agricultural policy options to promote the transition to climate-smart agriculture at the smallholder level. The policy context of this module is structured around six topics.

13.3.2 Sub-module Learning Outcomes

By the end of this module training, the following should be achieved:

- The role of agricultural policy frameworks in Kenya appreciated.
- Climate-smart agriculture practices, policy options and approaches identified and shared.
- Climate-smart-sensitive policy cycle mapped.
- Implementation of the climate-smart-sensitive policy at the county level explored and shared.
- Financing and Investments for Climate-smart Agriculture discussed and appreciated.
- The need for a Technology Policy explored and explained.

13.3.3 Sub-module Target Group

This module is intended for service providers, policy makers, public extension agents and lead farmers in the design and implementation of effective, climate-smart-sensitive agricultural policies.

13.3.4 Sub-module Users

This module is intended for use by master trainers who are members of the Core Team of Trainers (CTT). The trainers using this module should thoroughly familiarize themselves with the required participants handouts.

13.3.5 Sub-module Duration

The Module is estimated to take a minimum of 2 hours.

13.3.6 Sub-module Summary

Sub-module 13.3: Climate-Smart Agricultural Policy Options			
Sessions	**Training methods**	**Training materials**	**Time**
13.3.6.1 Introduction, learning expectations and outcomes	• Personal introductions • Group discussions • Plenary discussions • Presentations	• Flips charts • Felt pens • Projector • Laptop	15 minutes
13.3.6.2 Agricultural Policy Frameworks in Kenya	• Presentations • Practical exercises • Plenary discussions	• Flips charts • Felt pens • Projector • Laptop	15 minutes
13.3.6.3 Climate-smart agriculture practices, policy options and approaches	• Presentations • Practical exercises • Plenary discussions	• Flips charts • Felt pens • Projector • Participants Handouts • Laptop	15 minutes
13.3.6.4 Climate-smart-sensitive policy cycle	• Presentations • Plenary discussions	• Flips charts • Felt pens • PowerPoint Presentations • Participants Handouts • Laptop	15 minutes

13.3.6.5 Implementation of the climate-smart-sensitive policy at the county level	• Presentations • Practical exercise • Plenary discussions	• Flips charts • Felt pens • ProjectorParticipants Handouts • Laptop	15 minutes
13.3.6.6 Financing and Investments for Climate-smart Agriculture	• Presentations • Practical exercise • Plenary discussions	• Flips charts • Felt pens • Projector • Participants Handouts • Laptop	15 minutes
13.3.6.7 Technology Policy	• Presentations • Plenary discussions	• Flips charts • Felt pens • Projector • Participants Handouts • Laptop	15 minutes
13.3.6.8 Module Review	• Plenary discussion	Flip charts Felt pens	15 minutes
Total			2 hours

13.3.7 Facilitator's Guidelines

Module 13: Climate-Smart Agricultural Policy Options	
13.3.7.1 Introduction, Expectations and Outcomes (15 minutes)	**Session Guide**
(The facilitator welcomes trainees to the module on Climate-Smart Agricultural Policy Options and the importance in Agriculture to undergo a significant transformation in order to meet the related challenges of food security and climate change for enhancing employment opportunities, better incomes and livelihoods. They are then invited to introduce themselves and state their expectations). **Participants Expectations (7 minutes)** *(The facilitator requests the trainees to form groups and list their expectations).* **Module Outcomes (8 minutes)** *(The facilitator presents module objectives).* By the end of this module, trainee should be able to: • Appreciate the role of agricultural policy frameworks in Kenya. • Identify and share climate-smart agriculture practices, options and approaches. • Map the stages in climate-smart-sensitive policy cycle. • Explore the phases in the implementation of the climate-smart-sensitive policy at the county level. • Evaluate and select financing and investments options for Climate-smart Agriculture. • Explore and explain the need of a technology policy	• Summarize trainees' "Expectations" and display. • PowerPoint Presentation • Distribute Participants Handouts on Module Objectives and Training Program

13.3.7.2 Agricultural Policy Frameworks in Kenya (15 minutes)	
(Facilitator introduces the sub module and welcomes trainees to state their expectations) **Plenary Presentation (7 minutes)** • The role of agricultural policy frameworks in Kenya **Practical Exercise and Plenary Discussions (8 minutes)** *(The facilitator requests the trainees to form groups and identify the gaps between agricultural policy frameworks and the existing agricultural policies)*	• PowerPoint presentation • Distribute participants handouts • Group Exercise
13.3.7.3 Climate-smart agriculture practices, policy options and approaches (15 minutes)	**Session guide**
Plenary Presentation (7 minutes) • Considerations for climate-smart production systems • Existing systems, practices and methods suitable for climate-smart agriculture • Institutional and policy options • Ensuring farmer organizations for market access • Gendered approach **Practical Exercise and Plenary Discussions (8 minutes)** *(The facilitator requests the participants to form groups and identify the existing climate-smart agriculture practices and the relevant policy options for implementation)*	• PowerPoint presentation • Distribute participants handouts • Group Exercise
13.3.7.4 Climate-smart-sensitive policy cycle (15 minutes)	**Session Guide**
Plenary Presentation (7 minutes) • Stages in the climate-smart-sensitive policy cycle **Plenary Discussions (8 minutes)**	• PowerPoint presentation • Distribute participants handouts
13.3.7.5 implementation of the climate-smart-sensitive policy at the county level (15 minutes)	**Session Guide**
Plenary Presentation (7 minutes) • Phases in the implementation of the climate-smart-sensitive policy at the county level **Practical exercise and Plenary Discussions (8 minutes)** *(The facilitator requests the trainees to form groups and develop a programme showing steps, activities and stakeholders for the implementation of climate-smart policies)*	• PowerPoint presentation • Distribute participants handouts • Practical Exercise

13.3.7.6 Policy financing and investments for Climate-smart Agriculture (15 minutes)	Session Guide
Plenary Presentation (7 minutes) • Why financing is needed. • Financing gaps. • Sources of financing. • Financing mechanisms. • Connecting action to financing. • Types of subsidies to farmers. **Practical exercises and Plenary Discussions (8 minutes)** *(The facilitator requests the trainees to form groups and identify potential sources of financing, financing mechanisms and connecting action to financing)*	• PowerPoint presentation • Distribute participants handouts • Practical Exercise
13.7.7 Need of Technology Policy (15 minutes)	**Session guide**
Plenary Presentation (7 minutes) • What is a technology policy? • Why do we need technology policy? • Is technology policy inconsistent with a market oriented economy? • Technology policy in Kenya. **Plenary Discussions (8 minutes)**	• PowerPoint presentation • Distribute participants handouts
13.7.8 Module review 15 minutes)	**Session guide**
(The facilitator leads the trainees in reviewing the module) • Summarize the main points of the training and together with the participants review the main points. • Participant lists the main points leant during the training. • Discuss with participants new things learnt from this Module. • Ask the trainees what are some of the problems and issues that they have become more aware of in the module.	• Q& A session • Recap the main points • Test understanding • Participatory evaluation of the session

13.3.8 Participant's Handouts (Training and Reference Materials)

1. Alila, P.O. & Atieno, R. (2006). A paper for the Future Agricultures Consortium workshop, Institute of Development Studies, 20-22 March 2006. Future Agricultures.
2. Chronic Poverty Advisory Network (2012). Agriculture Policy Guide 2. Meeting the challenge of a new Pro-poor agricultural paradigm: The role of agricultural policies and programmes. www.chronicpovertynetwork.org
3. Chirwa, E.; Dorward, A.; Kathule, R.; Kumwenda, I.; Kydd, J., Poole, N.; Poulton, C. & Stockbridge, M. (undated). Farmer Organization for market access: Principles for policy and practice. Imperial College London. University

of Malawi. Agricultural Policy Research Unit. http://www.imperial.ac.uk/agriculturalsciences/research/sections/aebm/projects/farmerorg.htm

4. Food and Agriculture Organization of the United Nations (2016). The Gender in Agricultural Policies Analysis Tool (GAPo). FAO 2016.16274EN/2/01.18
5. Food and Agriculture Organization of the United Nations (FAO) (2010). "Climate-Smart" Agriculture. Policies, Practices and Financing for Food Security, Adaptation and Mitigation.
6. Ha-Joon Chang (2002). African Technology Policy Studies Network (ATPS). Who needs Technology Policy? Published by The African Technology Policy Studies Network, Nairobi, Kenya. ISBN: 9966-916-18-0
7. RoK (2007). Kenya adopted Vision 2030.
8. RoK (2010). Kenya Constitution

ANNEX 1: TRAINING PROGRAM

The training program presented here assumes that the trainees report on Sunday evening as the first day and leave 10 days later on Friday evening or Saturday morning.

Time	Day 0 (Sunday)-Travelling day	Duration	Remarks / Facilitator
Late Evening	• Arrival of participants and registration – Host • Setting up and prepare training venue and materials – CTT	2 Hours	The training venue and materials are ready for use
Time	**Day 1 (Monday)**	**Duration**	**Remarks / Facilitator**
8.00am-9.30am	**Session 1: Introduction, objectives & expectations** • Welcome by host and Prayers • Self-introductions –(CTT) • Introduction to KCSAP project • Official opening Ceremony (CEC) • Introduction to the training program (CTT)	10 minutes 20 minutes 20 minutes 20 minutes 20 minutes	The trainees relax and climate set for the ten-day training
9.30am-11.00am	**Session 2: Understanding importance of banana in Kenya's economy** • Banana as a climate smart crop • Banana situation in Kenya • Production and yield statistics	30 minutes 30 minutes 30 minutes	
11.00am-11.30am	**Tea Break (Group Photo)**	**30 minutes**	**Health Break**

11.30am-1.30pm	**Session 3: Banana production niche and appropriate** • **c**limatic conditions optimal yield Present a map showing banana production zones with their rainfall amount, temperature and soils • What are the climatic conditions, best soils for its optimal yield • Questions and answers	30 minutes 30 minutes 1 hour	
1.30pm-2.30pm	**Lunch Break**	**1 hour**	
2.30pm-4.30pm	**Session 4: Module review** • Participants review the module , summarize and display the main points to be considered when growing banana they have learnt • Discuss any issues and concerns from the participants	30 minutes 30 minutes	
4.30pm-5.00pm	**Coffee Break**	**30 minutes**	
	Close of Day 1		
Time	**Day 2 (Tuesday)**	**Duration**	**Remarks / Facilitator**
	Session 5: Introduction, objectives & expectations		
8.00am-9.00am	▪ Opening prayer and leveling of expectations ▪ Presentation of module's objectives		

9.00am-11.00am	**Session 6: Banana improved and their attributes** • Which banana are varieties are popular in the area and the reason • Available banana varieties and their attributes • Why improved varieties and criteria for selecting varieties	1 hour 30 minutes 30 minutes	
11.00am-11.30am	**Tea Break**	**30 minutes**	**Health Break**
11.30am-1.30pm	(a) **Session 7: Seed rate for appropriate plant density** (b) What quantity of banana do you use by acre? (c) What is the required plant population per acre/ha? (d) Calculating seed rate for different banana varieties (e) Discussions: Questions and answer	10 minutes 20 minutes 1 hour 30 minutes	
1.30pm-2.30pm	**Lunch Break**	**1 hour**	
2.30pm-4.30pm	**Session 8: Module review and group exercise** ▪ Participants to summarize the main points on a flip chart ▪ Practical exercise on estimation of seed rate ▪ Plenary discussions	10 minute 20 minutes 30 minutes	
4.30pm-5.30pm	**Coffee Break**	**30 minutes**	
Close of Day 2			

Time	Day 3 (Wednesday)	Duration	Remarks / Facilitator
	Session 9: Introduction, objectives & expectations		
8.00am-9.00am	▪ Opening prayer and leveling of expectations ▪ Presentation of module's objectives	30 minutes 30 minutes	
9.00am-11.00am	**Session 10: Banana systems in Kenya** • Main banana seed systems in Kenya ▪ Certified seed and importance of certified seed in banana production ▪ Seed quality and variety purity ▪ Quality seed production ▪ Plenary session	30 minutes 30 minutes 15 minutes 30 minutes 15 minutes	
10.30am -11.00am	**Tea Break**	**30 minutes**	**Health Break**
11.00am-1.00pm	**Session 11: Quality declared seed and requirements for production of banana quality declared seed** • Overview of quality declared seed. • Procedure for producing banana quality declared seed • Discussion	**45 minutes** **45 minutes** **30 minutes**	
1.00pm-2.00pm	**Lunch Break**	**1 hour**	
2.00pm-3.00pm	**Session 12: Practical on testing seed viability** (setting the experiment on seed germination text-planting of certified seed and banana grain from the market to test for germination test) ▪ Plenary session	30 minutes 30 minutes	

3.00pm-5.00pm	**Session 13: Module review** ▪ Conclusions and review of Module 3	1 hour	
4.30pm-5.00pm	**Coffee Break**	**30 minutes**	
	Close of Day 3		
Time	**Day 4 (Thursday)**	**Duration**	**Remarks / Facilitator**
	Session 14: Introduction, objectives & expectations		
8.00am-9.00am	▪ Opening prayer, Introduction and levelling of expectations ▪ Presentation of module objectives banana agronomic management practices and outcomes	30 minutes 30 minutes	
9.00am-11.00am	**Session 15: Banana GAPs for optimal banana yield** ▪ What are GAPs? ▪ Recommended GAPs ▪ Field crop management for optimal yield ▪ Plenary session	10 minutes 30 minutes 30minutes 20 minutes	
11.00am-11.30am	**Tea Break**	**30 minutes**	**Health Break**
11.30am-1.30pm	**Session 16: Appropriate inputs for banana optimal production and their correct dose** • What inputs do you use in banana production? • Appropriate inputs and levels for banana optimal production • Seed liming in moisture stresses conditions • Recap and plenary session	20 minutes 40 minutes 30 minutes 30 minutes	
1.30pm-2.30pm	**Lunch Break**	**1 hour**	

2.30pm-4.30pm	**Session 17: Tour demonstration plot for practical aspects on banana GAPs** (The participants will plant banana seed with recommended GAPs and using farmer farming method for comparison)	2 hours	
4.30pm-5.30pm	**Session 18: Module review** ▪ Summary of main points of the training; New knowledge acquired and Conclusions Module of 4	1 hour	
5.00pm-5.30pm	**Coffee Break**	**30 minutes**	
	Close of Day 4		
Time	**Day 5 (Friday)**	**Duration**	**Remarks / Facilitator**
	Session 19: Introduction, objectives & expectations		
8.30am-9.30am	▪ Introduction and levelling of expectations ▪ Presentation of module objectives on crop health and outcomes	30 minutes 30 minutes	
9.30am-10.30am	**Session 20: Major banana pests** ▪ Major pests of economic importance and yield losses caused by the pests ▪ Plenary session	30 minutes 30 minutes	
10.30am-11.00am	**Tea Break**	**30 minutes**	**Health Break**
11.00am-1.00pm	**Session 21: Banana Integrated Pest Management** • Participants share experiences on the control methods • Facilitator to presents the IPM options • Plenary discussions	30 minutes 1 hour 30 minutes	

1.00pm-2.00pm	**Lunch Break**	**1 hour**	
2.00pm-3.00pm	**Session 22: Banana Disease** • Major diseases of economic importance and their symptoms, yield losses and conditions that favor their development • Plenary session	30 minutes 30 minutes	
41.00pm-4.30pm	**Tea Break**	**30 minutes**	
3.00pm-5.00pm	**Session 23: Banana Integrated Disease Management (IDM)** ▪ Participants to share their experiences on the disease control used ▪ Facilitator to present the recommended IDM options ▪ Plenary session	30 minutes 1 hour 30 minutes	
5.00pm-6.00pm	**Session 24: Module review** ▪ Summary of main points of the training; ▪ New knowledge acquired and Conclusions Module on crop health ▪ Participatory evaluation of the session	20 minutes 20 minutes 20 minutes	
6.00pm-6.30pm	**Coffee Break**	**30 minutes**	
	Close of Day 5		
Time	**Day 6 (Monday)**	**Duration**	**Remarks / Facilitator**
	Session 25: Introduction, objectives & expectations		

8.00am-9.00am	▪ Introduction and levelling of expectations ▪ Presentation of module objectives and outcomes	30 minutes 30 minutes	
9.00am-11.00am	**Session 26: Nutrients requirement for optimal banana production** • What are the major soil fertility challenges in banana production • Nutrients requirement for the banana • Symptoms of nutrients deficiencies and the cause • Discussion	30 minutes 30 minutes 30 minutes 30minutes	
11.30am-12.00pm	**Tea Break**	**30 minutes**	**Health Break**
12.00pm-2.00pm	**Session 27: Soil sampling and analysis** • What is soil sampling • How to do soil sampling • Soil analysis results interpretation and recommendations • Participants visits a nearby demonstration plot for soil sampling exercise • Discussion	10 minutes 20 minutes 15 minutes 1 hour 15 minutes	
2.00pm-3.00pm	**Lunch Break**	**1 hour**	
3.00pm-5.00pm	**Session 28: Soil management techniques** • Participants share experiences on how they manage soil fertility in their farms • Facilitator presents soil management techniques for managing soil fertility challenges • Discussions	20minutes 1 hour 40 minutes	

5.00pm-6.00pm	**Session 29: Module review** ▪ Summary of main points of the training; New knowledge acquired and Conclusions Module of 6 ▪ Participatory evaluation of the session	30 minutes 30 minutes	
6.00pm-6.30pm	**Coffee Break**	**30 minutes**	
Time	**Day 7 (Tuesday)**	**Duration**	**Remarks / Facilitator**
	Session 30: Introduction, objectives & expectations		
8.00am-9.00am	▪ Introduction and levelling of expectations ▪ Presentation of module objectives and outcomes	30 minutes 30 minutes	
9.00am-11.00am	**Session 31: Role of banana as a food and nutrition security crop** • Nutritional and malnutrition challenges in Kenya and their associated symptoms • Facilitator shares the role of banana as a food and nutritional security crop • Discussion	45 minutes 45 minutes 30 minutes	
11.00am-11.30am	**Tea Break**	**30 minutes**	**Health Break**
11.30am-1.30pm	**Session 32: Banana nutritional value** **Banana nutritional composition and their role in human health** • Overview of the documented banana nutritional composition and their role in human health • Nutrient guidelines • Discussion	45 minutes 45 minutes 30 minutes	

1.30pm-2.30pm	**Lunch Break**	**1 hour**	
2.30pm-4.30pm	**Session 33: Banana-based value added products** • Participants share experiences on banana utilization at the household level • Facilitator shares the role of banana as a food and nutritional security crop • Practical on making various banana-based value-added products	15 minutes 15 minutes 1 hour 30 minutes	
4.30pm-5.30pm	**Session 34: Module review** ▪ Participants summarize the main points of the training; New information acquired ▪ Participatory evaluation of the session	1 hour	
5.30pm-6.00pm	**Coffee Break**	**30 minutes**	
Time	**Day 8 (Wednesday)**	**Duration**	**Remarks / Facilitator**
8.00am–9.00am	**Session 35: Introduction, objectives & expectations** ▪ Introduction and levelling of expectations ▪ Presentation of module objectives and outcomes	30 minutes 30 minutes	
9.00am–11.00am	**Session 36 Banana harvesting and post-harvest handling** ▪ Overview of the banana harvesting methods ▪ Use of mechanization –banana thresher ▪ Crop production guidelines ▪ Discussion	30 minutes 30 minutes 30 minutes 30 minutes	
11.00am–11.30am	**Tea Break**	**30 minutes**	**Health Break**

11.30am–1.30pm	**Session 37**: **Banana post-harvest losses and the economic importance** • Banana post-harvest losses and their economic implications in banana production, consumption and marketing • Discussion	1 hour 1 hour	
1.30pm–2.30pm	**Lunch Break**	**1 hour**	
2.30pm–4.30pm	**Session 38: Post-harvest management practices** • Participants share experiences on the post-harvest control methods especially the ITK • Facilitators shares the sustainable post-harvest management practices • Practical demonstration on post-harvest management • Discussion to identify the sustainable and gender friendly management option	10 minutes 20 minutes 1 hour 30 minutes	
4.30pm–5.00pm	**Session 39: Module review** ▪ Summary of main points of the training; New knowledge acquired and Conclusions ▪ Discussion and participatory evaluation of the session	30 minutes 30 minutes	
5.00pm-5.30pm	**Coffee Break**	**30 minutes**	
Time	**Day 9 (Thursday)**	**Duration**	**Remarks / Facilitator**

8.00am–9.00am	**Session 40: Introduction, objectives & expectations**		
	▪ Introduction and levelling of expectations ▪ Presentation of module objectives and outcomes	30 minutes 30 minutes	
9.00am–10.30am	**Session 41: Banana as a business** ▪ Importance of banana production as a business; ▪ SWOT analysis	30 minutes 1 hour	
10.30am–11.00am	**Tea Break**	**30 minutes**	**Health Break**
11.00am–12.30pm	**Session 42: Opportunities for banana products in local and international markets** • Participants shares experiences in banana marketing • Requirements for participation in domestic, regional and international markets • Facilitator presents diverse banana marketable products and their pricing	30 minutes 30 minutes 30 minutes	
12.30pm–2.00pm	**Session 43: Market analysis** • Facilitator shares banana target market, market trend and standards • Understanding consumer preference • Marketing strategy (product packaging/ promotion)	30 minutes 30 minutes 30 minutes	
2.00pm–3.00pm	**Lunch Break**	**1 hour**	

3.00pm–5.00pm	**Session 44: Business plan** ▪ Definition of a business plan and its components; ▪ Objective of a business plan; ▪ Requirement of operating a business plan ▪ Practical exercises to develop a business plan ▪ Plenary session and practical exercises	10 minutes 20 minutes 30 minutes 30 minutes 30 minutes	
5.00pm–6.00pm	**Session 45: Module review** ▪ Summary of main points of the training on a flip chart; New information gained on banana farming as a business ▪ Participatory evaluation of the session	30minutes 30 minutes	
6.00pm–6.30pm	**Coffee Break**	**30 minutes**	
Time	**Day 10 (Friday)**	**Duration**	**Remarks / Facilitator**
	Session 46: Introduction, objectives & expectations		
8.00am–9.00am	▪ Introduction and levelling of expectations ▪ Presentation of module objectives and outcomes	30 minutes 30 minutes	

9.00am–10.30am	**Session 47: Gender mainstreaming and social inclusion in banana value chain** ▪ Definition of gender ▪ What is gender mainstreaming and why it is important. ▪ Who does what? (gender division of roles in bananas value chain) ▪ Who owns what? (access and control of resources & benefits) ▪ Who makes which decisions? ▪ Socio-cultural limitations related to banana value chain ▪ Existing policies in support of gender mainstreaming	30 minutes 30 minutes	
10.30am-11.00am	**Tea Break**	**30 minutes**	**Health Break**
11.30am – 12.30	Group work	30 minutes	
12.30 -1:00	General discussions	30 minutes	
1:00 – 2:00pm	**Lunch break**		

2:00pm - 2.30 pm	**Session 48: Youth empowerment in banana value chain** • Why agriculture is not attractive to youth • Youth's role in the value chain • Strategies to empower youth in banana value chain Group work General discussions	30 minutes 30 minutes 15 minutes	
2.30pm-3.00pm	**Session 49: Women empowerment in banana value chain** • Women's role in the value chain • Challenges facing women in the value chain • Strategies for empowering women in the value chain Group work General discussions	30 minutes 30 minutes 30 minutes 15 minutes	
3.00pm-3:30pm	**Tea Break**	**30 minutes**	

3:30pm -4:15pm	**Session 50: Strategies for inclusion of vulnerable and marginalized groups in banana value chain (1:15 minutes)** • Who are vulnerable and marginalized groups (VMGs) • Why gender inequality exist • Social inclusion and why • Strategies of inclusion of VMGs Group work General discussions	30minute 30 minutes	
4:15pm – 5:30 pm	**Session 50: Environmental and social management framework (ESMF) (1:15 minutes)** • Objective of ESMF in banana value chain • Environmental, social risks • Environmental and social safeguards • Environmental and socioeconomic impacts of banana value chain activities Group exercise discussions	30 minutes 30 minutes 15 minutes	

5:30 – 6:30 pm	**Session 50: Module review** ▪ Summary of main points of the training; New knowledge acquired and Conclusions ▪ Participatory evaluation of the session	30 minutes 30 minutes	

Annexe 2: General Reference Materials

Category / Modules	Publication title	Reference types	No Pages	Farmer Category A= New entrant/Banana farmer B= Established BANANA Farmer
1. General BANANA Farming Manuals				
	A guide to growing bananas in Eastern African Highlands	Training manual	40	A/B
	Proceedings of the International Conference On Banana and Plantain In Africa: Harnessing International Partnerships to Increase Research Impact	Book	430	A/B
IMPROVED BANANA VARIETIES	Banana Production Field Technical Guide	Training manual	294	A/B
	Tissue culture	Leaflet	6	A/B
	KALRO New Bananas	Digital Platform	9	A/B
		Book	68	A/B
PROPAGATION	How to multiply clean banana planting material from existing plants	Leaflet	6	A/B

	Jinzi ya kuzalisha mbegu ya ndizi kutokana na mimea iliyoko shambani	Leaflet	6	A/B
Soil Fertility and water Management for Banana Farmers				
Banana diseases and their control	Grow Panama disease resistant Apple banana	Leaflet	6	A/B
	Kuza ndizi zinazohimili ugonjwa wa "Black Sigatoka"			
	Management of Banana Xanthomonas Wilt (BXW) in Kenya	Leaflet	6	A/B
	Randy C. Ploetz, 2015. Fusarium wilt of banana. Phytopathology Review. 23 Nov 2015https://doi.org/10.1094/PHYTO-04-15-0101-RVW			
	Randy C. Ploetz, Gert H.J. Kema, and Li-Jun Ma, 2015. Impact of Diseases on Export and Smallholder Production of Banana. Annual Review of Phytopathology. Vol. 53:269-288 https://doi.org/10.1146/annurev-phyto-080614-120305			
	Banana Diseases and their control. Agropedia. Reviewed online on 21st June 2019. http://agropedia.iitk.ac.in/content/banana-diseases-their-control			

	S.P. Raut and Suvarna Ranade, 2004. Diseases of Banana and their Management Diseases of Fruits and Vegetables: Volume II pp 37-52. https://link.springer.com/chapter/10.1007/1-4020-2607-2_2			
Banana insect pests and their control	7 1. Banana Weevil. Infonet Biovision. https://www.infonet-biovision.org/PlantHealth/Pests/Banana-weevil			
Postharvest handling and management	3. Mohapatra, D., Mishra, S., Singh, C.B. and Jayas, D.S., 2011. Post-harvest processing of banana: opportunities and challenges. *Food and Bioprocess Technology*, *4*(3), pp.327-339. 4. Anyasi, T.A., Jideani, A.I. and Mchau, G.R., 2013. Functional properties and postharvest utilization of commercial and noncommercial banana cultivars. *Comprehensive Reviews in Food Science and Food Safety*, *12*(5), pp.509-522.			
			24	B
BANANA Business	1. Banana Agribusiness: How it works. You tube video https://www.youtube.com/watch?v=oRr6n1pw9Ik			B/A
			2	A

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